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Uni-Perceiver v2: A Generalist Model for Large-Scale Vision and Vision-Language Tasks

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Abstract

Despite the remarkable success of foundation models, their task-specific fine-tuning paradigm makes them inconsistent with the goal of general perception modeling. The key to eliminating this inconsistency is to use generalist models for general task modeling. However, existing attempts at generalist models are inadequate in both versatility and performance. In this paper, we propose Uni-Perceiver v2, which is the first generalist model capable of handling major large-scale vision and vision-language tasks with competitive performance. Specifically, images are encoded as general region proposals, while texts are encoded via a Transformer-based language model. The encoded representations are transformed by a task-agnostic decoder. Different tasks are formulated as a unified maximum likelihood estimation problem. We further propose an improved optimizer to ensure stable multi-task learning with an unmixed sampling strategy, which is helpful for tasks requiring large batch-size training. After being jointly trained on various tasks, Uni-Perceiver v2 is capable of directly handling downstream tasks without any task-specific adaptation. Results show that Uni-Perceiver v2 outperforms all existing generalist models in both versatility and performance. Meanwhile, compared with the commonly-recognized strong baselines that require tasks-specific fine-tuning, Uni-Perceiver v2 achieves competitive performance on a broad range of vision and vision-language tasks.

1. Introduction

Learning a *general perception model* that can handle various modalities and tasks is widely regarded as an important step towards artificial general intelligence. Due to its difficulty, many works (*e.g.,* Florence [45], CoCa [44], BEiT-3 [40]), also known as *foundation models* [3], instead focus on a fallback solution of learning a general representation encoder that can be adapted (*e.g.,* fine-tuned) to various downstream tasks. By performing large-scale pre-training on massive multi-modal task-agnostic data, these works have demonstrated the superiority by pushing the state-of-the-art results on a broad range of tasks including single-modal tasks (*e.g.,* image classification and object detection) and also cross-modal tasks (*e.g.,* image captioning and image retrieval).

Despite the success, there is still a considerable gap between foundation models and the goal of general perception modeling. While foundation models only focus on general representation learning, task modeling is neglected. Traditional task-specific fine-tuning paradigm is still utilized (see Fig. 1). This significantly increases the marginal cost of adapting pre-trained models to various downstream tasks, making it difficult to meet the rapidly growing demands of diverse downstream tasks and scenarios. Such a task-specific fine-tuning paradigm of foundation models is inconsistent with the goal of general perception modeling.

Instead of performing task-specific fine-tuning, generalist models process different tasks with shared architecture and parameters, which is aligned with the goal of general perception modeling. It not only reduces the cost of handling diverse tasks but also enables task collaboration. Most existing attempts on generalist models are sequence-

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Uni-Perceiver v2：面向大规模视觉和视觉-语言任务的通用模型

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摘要

尽管基础模型取得了显著成功，但其面向特定任务的微调范式使其与通用感知建模的目标不一致。消除这种不一致性的关键在于使用通才模型进行通用任务建模。然而，现有关于通才模型的尝试在多功能性和性能方面都仍显不足。本文提出 Uni-Perceiver v2，这是首个能够以具有竞争力的性能处理主要大规模视觉和视觉-语言任务的通才模型。具体而言，图像被编码为通用区域提议，而文本则通过基于 Transformer 的语言模型进行编码。编码后的表示由一个任务无关的解码器进行转换。不同任务被表述为统一的最大似然估计问题。我们进一步提出了一种改进的优化器，通过非混合采样策略确保稳定的多任务学习，这对于需要大批量训练的任务很有帮助。在经过多种任务的联合训练后，Uni-Perceiver v2 能够在无需任何特定任务适配的情况下直接处理下游任务。结果表明，Uni-Perceiver v2 在多功能性和性能方面均优于所有现有通才模型。同时，与通常被认可的、需要特定任务微调的强基线相比，Uni-Perceiver v2 在广泛的视觉和视觉-语言任务上取得了具有竞争力的性能。

1. 引言

学习一个能够处理多种模态和任务的通用感知模型，被广泛认为是迈向通用人工智能的重要一步。由于其难度较大，许多工作（例如 Florence [45]、CoCa [44]、BEiT-3 [40]），也被称为基础模型 [3]，转而关注一种折中方案：学习一个通用表示编码器，使其能够适配（例如微调）到各种下游任务。通过在大规模、多模态、任务无关的数据上进行大规模预训练，这些工作已通过在广泛任务上推动最先进结果的发展而展示出其优越性，这些任务包括单模态任务（例如图像分类和目标检测）以及跨模态任务（例如图像描述生成和图像检索）。

尽管取得了成功，基础模型与通用感知建模的目标之间仍存在相当大的差距。基础模型仅关注通用表征学习，而忽略了任务建模。传统的特定任务微调范式仍在被采用（见图 1）。这显著增加了将预训练模型适配到各种下游任务的边际成本，使其难以满足多样化下游任务和场景快速增长的需求。基础模型这种特定任务微调范式与通用感知建模的目标并不一致。

通用模型并不执行特定任务的微调，而是使用共享的架构和参数处理不同任务，这与通用感知建模的目标一致。它不仅降低了处理多样化任务的成本，还能实现任务协作。现有大多数关于通用模型的尝试都是基于序列的-

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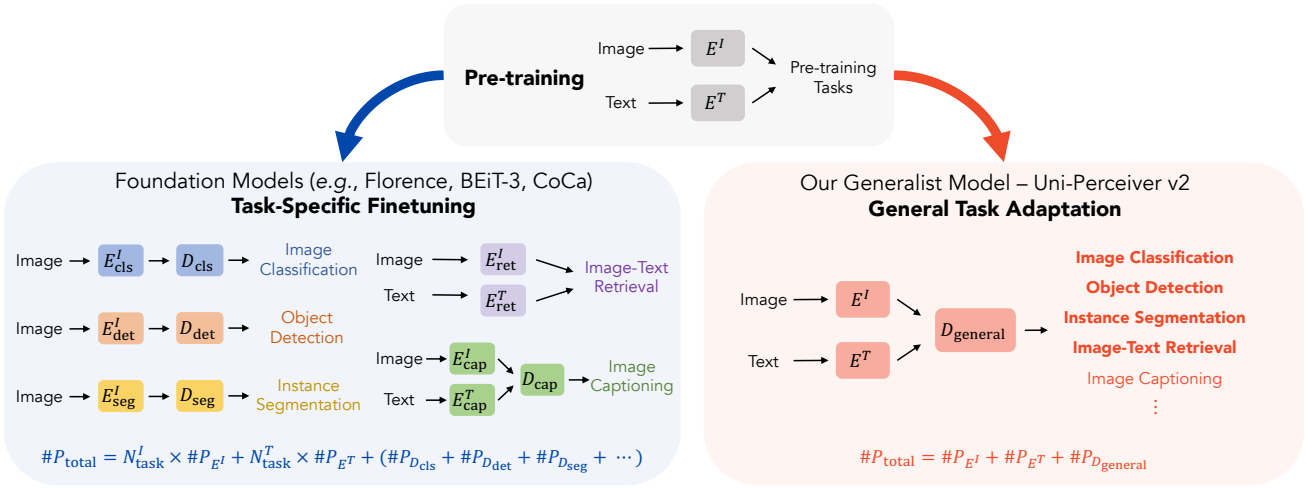


Figure 1. Comparison of foundation models and Uni-Perceiver v2. E^I and E^T denote the image encoder and text encoder, respectively. In existing foundation models, task-specific decoders $D_{\text{cls}}, D_{\text{det}}, \ldots$ are employed to tune E^I and E^T in different task-specific finetuning. The total number of parameters $\#P_{\text{total}}$ in adaptation grow with the number of visual/linguistic tasks, denoted as N^I_{task} and N^T_{task}, respectively. By contrast, our Uni-Perceiver v2 shares all parameters across various downstream tasks with a general decoder D_{general}, where no task-specific fine-tuning is incorporated. Better than previous generalist models, our method can also effectively handle pillar tasks such as image classification, object detection, instance segmentation, and image-text retrieval.

to-sequence (seq2seq) models [2, 6, 11, 15, 23, 29, 39, 43]. However, these attempts are inadequate in both versatility and performance: (1) some pillar vision and vision-language tasks as listed in Tab. 1 cannot be handled, *e.g.,* image-text retrieval, object detection, and instance segmentation; (2) the accuracy and inference speed still lag significantly behind state-of-the-art task-specific methods. Another line of research named Uni-Perceivers [1, 48] builds generalist models supporting both generation and non-generation tasks. Nevertheless, they still cannot handle many vital tasks such as detection and segmentation.

To develop generalist models with better versatility and performance, our core idea is to encode images as general region proposals consisting of the semantic, bounding box and segmentation mask representations. Compared with previous methods where images are represented as non-overlapping patches, this design makes our localization modeling more expressive and flexible. This explicit utilization of localization clues not only greatly reduces the difficulty of handling localization tasks such as image detection and segmentation, but also provides richer features for non-localization tasks, thus enabling more general task modeling and better performance.

In this paper, we propose Uni-Perceiver v2 as a generalist model capable of handling major large-scale vision and vision-language tasks as listed in Tab. 1. Specifically, images are encoded as a concatenation of global and regional representations via a region proposal network, while texts are encoded via a Transformer-based language model. Both the image and text encoders can benefit from off-the-shelf pre-trained models, which reduces the demand for training data and resources and ensures performance. The encoded representations are transformed by a shared modality-agnostic Transformer [36] network to obtain the decoded representations. Following Uni-Perceivers [1, 48], different tasks are formulated as a unified maximum likelihood estimation problem and are jointly learned to enable general task adaptation. We further propose an improved optimizer named MT-AdamW to ensure stable multi-task learning with an unmixed sampling strategy which only samples one task for all GPUs per iteration. This is very helpful for tasks requiring large batch size training.

Uni-Perceiver v2 is the first generalist model achieving competitive results on major large-scale vision and vision-language tasks including object detection, instance segmentation, image classification, image captioning, and image-text retrieval, except for image generation that has not been verified due to limited computational resources. After being jointly trained on various tasks, it can directly handle a broad range of tasks without any task-specific adaption, achieving state-of-the-art performance among existing generalist models. Our contributions are summarized as:

- We propose Uni-Perceiver v2, which is the first generalist model capable of handling both localization and non-localization tasks with competitive performance. The general region proposal encoding of images brings more flexible and expressive localization modeling.
- To improve the effectiveness of multi-task learning, we adopt an unmixed sampling strategy to enable large batch-size training and develop an improved optimizer named

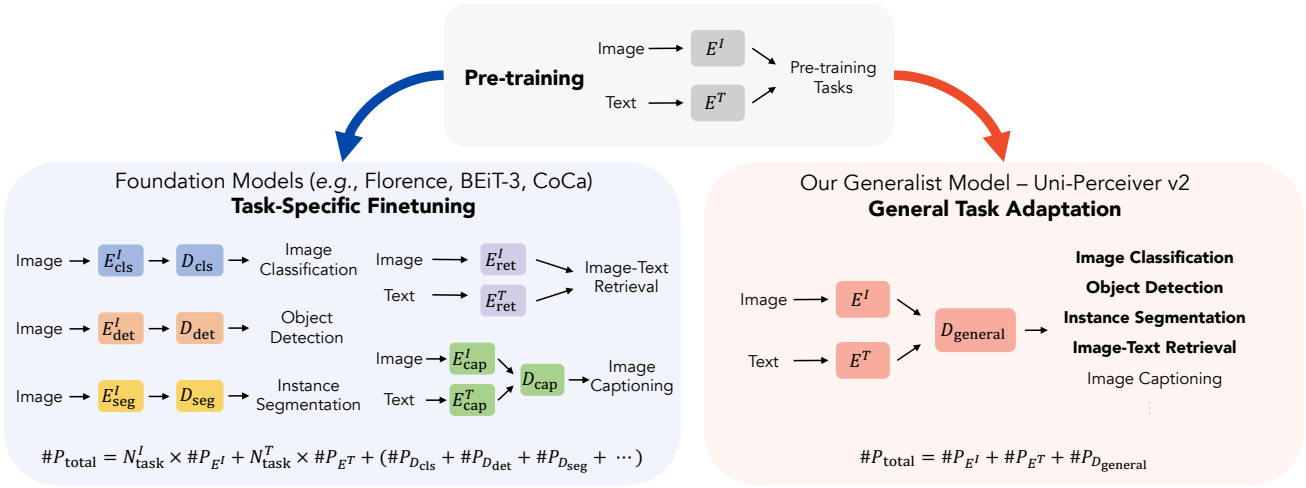


图 1. 基础模型与 Uni-Perceiver v2 的比较。E^I 和 E^T 分别表示图像编码器和文本编码器。在现有基础模型中，采用任务特定解码器 D_{cls}、$D_{\text{det}}, \ldots$ 在不同的任务特定微调中对 E^I 和 E^T 进行调优。自适应中的参数总数 $\#P_{\text{total}}$ 会随着视觉/语言任务数量的增加而增长，分别记为 N^I_{task} 和 N^T_{task}。相比之下，我们的 Uni-Perceiver v2 通过通用解码器 D_{general} 在各种下游任务之间共享所有参数，其中不包含任务特定的微调。优于以往的通才模型，我们的方法还能够有效处理图像分类、目标检测、实例分割和图像-文本检索等支柱任务。

到序列（seq2seq）模型 [2, 6, 11, 15, 23, 29, 39, 43]。然而，这些尝试在通用性和性能方面都存在不足：（1）如表 1 所列，一些核心的视觉和视觉-语言任务无法处理，例如图文检索、目标检测和实例分割；（2）其准确率和推理速度仍显著落后于最先进的任务专用方法。另一条名为 Uni-Perceivers [1, 48] 的研究路线构建了支持生成和非生成任务的通用模型。尽管如此，它们仍然无法处理许多重要任务，例如检测和分割。

为了开发具有更好通用性和性能的通用模型，我们的核心思想是将图像编码为通用区域提议，这些提议由语义、边界框和分割掩码表示组成。与以往将图像表示为不重叠图块的方法相比，这种设计使我们的定位建模更具表达能力和灵活性。对定位线索的这种显式利用，不仅大大降低了处理图像检测和分割等定位任务的难度，还为非定位任务提供了更丰富的特征，从而实现更通用的任务建模和更好的性能。

在本文中，我们提出 Uni-Perceiver v2，作为一种通才模型，能够处理表 1 中列出的主要大规模视觉和视觉-语言任务。具体而言，图像通过区域提议网络被编码为全局表示和区域表示的拼接，而文本则通过基于 Transformer 的语言模型进行编码。图像和文本编码器都可以受益于 off-

现成的预训练模型，这降低了对训练数据和资源的需求，并确保了性能。编码后的表示由一个共享的、模态无关的 Transformer [36] 网络进行转换，以获得解码后的表示。遵循 Uni-Perceivers [1, 48]，不同任务被表述为统一的最大似然估计问题，并进行联合学习，以实现通用的任务适配。我们进一步提出了一种改进的优化器，名为 MT-AdamW，以确保在未混合采样策略下进行稳定的多任务学习；该策略在每次迭代中仅为所有 GPU 采样一个任务。这对于需要大批量训练的任务非常有帮助。

Uni-Perceiver v2 是首个在主要大规模视觉及视觉-语言任务上取得有竞争力结果的通用模型，这些任务包括目标检测、实例分割、图像分类、图像描述生成以及图像-文本检索；但由于计算资源有限，图像生成尚未得到验证。在各种任务上进行联合训练后，它无需任何特定任务的适配即可直接处理广泛的任务，并在现有通用模型中达到最先进的性能。我们的贡献总结如下：

- 我们提出了 Uni-Perceiver v2，这是首个能够以具有竞争力的性能同时处理定位任务和非定位任务的通用模型。图像的通用区域提议编码带来了更灵活且更具表现力的定位建模。
- 为了提高多任务学习的有效性，我们采用了一种非混合采样策略，以实现大批量训练，并开发了一种改进的优化器，名为

Categories	Specific Tasks
Retrieval	**Image-text retrieval**
Classification	**Image classification** Region categorization Situation recognition
Localization	**Object detection** Key point detection Pose estimation Referring expression grounding Human object interaction Relation detection Optical character recognition Object localization
Mask Predication	**Instance segmentation** Semantic segmentation Panoptic segmentation
Image Generation	**Image synthesis** Image inpainting Segment-based image generation Style transferring Depth estimation Surface normal estimation Image infilling Image super resolution
Image to Text	**Image captioning** Visual question answering Region captioning Grounded VQA Grounded captioning Visual commonsense reasoning

Table 1. Categories of mainstream vision and vision-language tasks. Pillar tasks of different downstream task categories are in **bold**. These pillar tasks are the most representative tasks in each category, where other tasks can be derived from them. Uni-Perceiver v2 is able to effectively handle the underlined pillar tasks, except for image synthesis that has not been verified due to limited computational resources.

MT-AdamW to mitigate the instability in gradients.

- Uni-Perceiver v2 outperforms all existing generalist models in both versatility and performance. Without any task-specific adaption, Uni-Perceiver v2 achieves competitive performance on a broad range of downstream tasks compared with commonly-recognized strong baselines that require task-specific fine-tuning, demonstrating its strong ability of general task modeling.

2. Related Work

Foundation Vision Models are "designed to be adapted (*e.g.,* fine-tuned) to various downstream tasks by *pre-training* on broad data at scale" [3]. Such large-scale pre-trained vision models have shown effectiveness in enriching data encoding capacity, alleviating data hunger, and improving the performance of downstream tasks.

Image classification on ImageNet-1k [9] has been the mainstream pre-training paradigm for a long period. However, as the model size grows, larger annotated datasets are required to avoid over-fitting in pre-training, such as ImageNet-21k [9], Instagram-1B [24], JFT-300M [35] and JFT-3B [46]. Inspired by the success of linguistic pre-training on massive web-crawled text, CLIP [27] and ALIGN [13] have begun to focus on multi-modal contrastive pre-training on web-scale noisy image-text pairs to learn aligned image and text representations. SimVLM [41] employs the multi-modal sequence generation task for pre-training. FLAVA [34] combines contrastive and generative pre-training to handle both unimodal and multimodal tasks. UniCL [42] and CoCa [44] jointly use human-annotated and web-crawled data. Florence [45] and INTERN [31] increase the scale and diversity of pre-training data to enhance the representation capability. OmniVL [37] proposes to incorporate both image-language and video-language tasks in its pre-training. BEiT-3 [40] unifies pre-training objectives for different modalities as a single masked data modeling task, achieving state-of-the-art results on a wide range of downstream tasks.

These works on foundation models only focus on general representation learning, while neglecting task modeling. When adapting them to downstream tasks, the traditional task-specific fine-tuning paradigm is still utilized, which is inconsistent with the goal of general perception modeling. Meanwhile, with the rapidly growing demands of diverse tasks and scenarios, the task-specific fine-tuning paradigm would result in a prohibitive marginal cost for data collection, data annotation, model training, and model storage.

Generalist models handle various tasks with shared architecture and parameters, which have been long pursued by the machine learning community. Recently, inspire by the success of sequence-to-sequence (seq2seq) models in NLP field [28], OFA [39], Flamingo [2], and GIT [38] propose to model various tasks as a sequence generation task. Unified-IO [23], Pix2Seq v2 [6], and UniTab [43] further develop this method to support more tasks by introducing discrete coordinate tokens, thus location information can be encoded or decoded by the unified models. Beyond that, Gato [29] succeeds in unifying reinforcement learning tasks into the seq2seq framework. GPV [11] also builds a general-purpose vision system by adding a seq2seq module on a DETR [4]-based visual encoder.

However, these methods with seq2seq formulation are still inadequate in both versatility and performance: (1) They cannot handle some core vision tasks, *e.g.,* image-text retrieval, object detection, and instance segmentation. Although Pix2Seq v2 [6] includes detection and instance segmentation tasks, its performance and inference speed still lag significantly behind state-of-the-art task-specific meth-

Categories	Specific Tasks
Retrieval	**Image-text retrieval**
Classification	**Image classification** Region categorization Situation recognition
Localization	**Object detection** Key point detection Pose estimation Referring expression grounding Human object interaction Relation detection Optical character recognition Object localization
Mask Predication	**Instance segmentation** Semantic segmentation Panoptic segmentation
Image Generation	**Image synthesis** Image inpainting Segment-based image generation Style transferring Depth estimation Surface normal estimation Image infilling Image super resolution
Image to Text	**Image captioning** Visual question answering Region captioning Grounded VQA Grounded captioning Visual commonsense reasoning

表 1. 主流视觉和视觉-语言任务的类别。不同下游任务类别的支柱任务以粗体显示。这些支柱任务是各类别中最具代表性的任务，其他任务可由它们衍生而来。Uni-Perceiver v2 能够有效处理带下划线的支柱任务，但图像合成除外，由于计算资源有限，该任务尚未得到验证。

MT-AdamW 用于缓解梯度的不稳定性。

- Uni-Perceiver v2 在通用性和性能方面均优于所有现有的通用模型。无需任何特定任务的适配，Uni-Perceiver v2 在广泛的下游任务上取得了具有竞争力的性能，可与通常被认为强大的、需要特定任务微调的基线方法相媲美，展现出其强大的通用任务建模能力。

2. 相关工作

基础视觉模型被“设计为通过在大规模广泛数据上进行预训练，以适应（例如，微调）各种下游任务”[3]。这类大规模预训练视觉模型已被证明能够有效增强数据编码能力、缓解数据饥渴问题，并提升下游任务的性能。

在很长一段时间里，ImageNet-1k [9] 上的图像分类一直是主流的预训练范式。然而，随着模型规模的增长，需要更大的标注数据集来避免预训练中的过拟合，例如 ImageNet-21k [9]、Instagram-1B [24]、JFT-300M [35] 和 JFT-3B [46]。受大规模网络爬取文本上的语言预训练成功启发，CLIP [27] 和 ALIGN [13] 开始关注在网络规模的噪声图文对上进行多模态对比预训练，以学习对齐的图像和文本表示。SimVLM [41] 采用多模态序列生成任务进行预训练。FLAVA [34] 结合对比式和生成式预训练，以同时处理单模态和多模态任务。UniCL [42] 和 CoCa [44] 联合使用人工标注数据和网络爬取数据。Florence [45] 和 INTERN [31] 扩大预训练数据的规模和多样性，以增强表示能力。OmniVL [37] 提出在其预训练中同时融入图像-语言和视频-语言任务。BEiT-3 [40] 将不同模态的预训练目标统一为单一的掩码数据建模任务，并在广泛的下游任务上取得了最先进的结果。

这些关于基础模型的工作仅关注通用表征学习，而忽视了任务建模。在将它们适配到下游任务时，仍然采用传统的任务特定微调范式，这与通用感知建模的目标并不一致。同时，随着多样化任务和场景需求的快速增长，任务特定微调范式将导致数据收集、数据标注、模型训练和模型存储方面难以承受的边际成本。

通用模型使用共享的架构和参数来处理各种任务，这一直是机器学习社区长期追求的目标。最近，受 NLP 领域中序列到序列（seq2seq）模型成功的启发 [28]，OFA [39]、Flamingo [2] 和 GIT [38] 提出将各种任务建模为序列生成任务。Unified-IO [23]、Pix2Seq v2 [6] 和 UniTab [43] 通过引入离散坐标标记进一步发展了这一方法，以支持更多任务，从而使位置信息能够由统一模型进行编码或解码。除此之外，Gato [29] 成功地将强化学习任务统一到 seq2seq 框架中。GPV [11] 也通过在基于 DETR [4] 的视觉编码器上添加一个 seq2seq 模块，构建了一个通用视觉系统。

然而，这些采用 seq2seq 形式的方法在通用性和性能方面仍然不足：（1）它们无法处理一些核心视觉任务，例如图文检索、目标检测和实例分割。尽管 Pix2Seq v2 [6] 包含了检测和实例分割任务，但其性能和推理速度仍显著落后于最先进的任务专用方法。

ods [17, 47]; (2) The non-parallel auto-regressive decoding leads to slow inference speed. For example, image classification requires calculating and comparing the cumulative probabilities of all category names conditioned on the given image; (3) They also suffer from the task-interference issue in multi-task learning, resulting in performance degradation compared with task-specific models.

Alternatively, Uni-Perceivers [1, 48] formulate different tasks as finding the maximum likelihood target for each input through the representation similarity regardless of their modality, making it possible to support both generation and non-generation tasks. Nevertheless, they still cannot handle image detection and segmentation tasks.

3. Revisiting Uni-Perceivers

Unified Modeling of Perception Tasks. Uni-Perceiver [1] proposes to reformulate different tasks as a unified maximum likelihood estimation problem. Specifically, each task is defined with a set of inputs and a set of candidate targets from arbitrary combinations of modalities. The inputs and targets are first encoded with a modality-specific tokenizer with linear projection. Then the encoded representations are transformed by modality-agnostic decoder with shared parameters for different tasks. Given an input, the unified task objective is defined as finding the target with the maximum likelihood with the input.

Mitigating Task Interference. Multi-task learning with fully shared parameters could introduce interference between different tasks. Uni-Perceiver-MoE [48] proposes Conditional MoEs to address the task-interference issue. Specifically, for each input token, a routing decision is calculated depending on specific routing strategy, which sparsely activates a small portion of experts to process this token. The corresponding output of an input token is the linearly weighted combination of those selected experts by the routing decision. Conditional MoEs mitigate the interference issue by allowing conflicting modalities and tasks using separate parameters without introducing any task-specific modules.

Limitations. Although Uni-Perceivers aim to process different tasks with a unified architecture, it fails to handle detection and segmentation tasks due to the lack of localization information in its encoded features. Meanwhile, Uni-Perceivers do not integrate off-the-shelf encoder models, making it unable to benefit from existing large-scale pre-trained encoders. This potentially increases its demand for pre-training data and resources, limiting its performance.

4. Method

4.1. Encoding Images as General Region Proposals

Most existing generalist models [1, 48] represent images as non-overlapping patches with fixed sizes. This design is rather coarse and limited in modeling objects of varying sizes and shapes in images, making it difficult to handle localization tasks such as detection and segmentation.

In order to enable more expressive and flexible localization modeling, we propose to encode the input image as a sequence of general region proposals. Specifically, given an input image $x \in \mathbb{R}^{H\times W}$ with height H and width W, a network $f_{\text{image}}(\cdot)$ is employed to encode the image as the concatenation of global and regional representations as

$$f_{\text{image}}(x) = \text{Concat}\left(\{q_i^{\text{global}}\}_{i=1}^{M}\,,\ \{q_j^{\text{proposal}}\}_{j=1}^{N}\right), \quad (1)$$

where $q_i^{\text{global}} \in \mathbb{R}^d$ are the global representations of the whole image, and $q_j^{\text{proposal}} \in \mathbb{R}^d$ are the regional representations of candidate object proposals in the image.

Following the common practice in localization tasks, an image backbone network (*e.g.,* ResNet [12]) is firstly employed to extract the multi-scale feature maps $\{\mathcal{F}_l\}_{l=1}^{L}$, where L is the number of feature scales (*e.g.,* $L = 4$).

Regional Representations. A Transformer [36]-based region proposal network is applied on top of the multi-scale feature maps $\{\mathcal{F}_l\}_{l=1}^{L}$ to extract a set of O candidate object proposals $\{q_j^{\text{sem}}, q_j^{\text{box}}, q_j^{\text{mask}}\}_{j=1}^{O}$, where $q_j^{\text{sem}} \in \mathbb{R}^d$, $q_j^{\text{box}} \in \mathbb{R}^4$, and $q_j^{\text{mask}} \in \mathbb{R}^{H\times W}$ are the semantic, bounding box, and segmentation mask representations of the j-th proposal, respectively. The region proposal network is similar to MaskDINO [17], but only considers foreground-background binary classification. See Appendix for detailed implementation. These three representations are then fused as the regional representation as

$$q_j^{\text{proposal}} = q_j^{\text{sem}} + \mathcal{B}(q_j^{\text{box}}) + \mathcal{M}(q_j^{\text{mask}}), \quad (2)$$

where $\mathcal{B}$ denotes the positional encoding of box coordinates. $\mathcal{M}$ uses adaptive average pooling to scale the mask predictions to the size of 28×28. Both $\mathcal{B}$ and $\mathcal{M}$ are followed by linear projections to match the feature dimension.

Global Representations. The global representations are extracted from the last-scale feature map $\mathcal{F}_L \in \mathbb{R}^{h\times w}$ with height h and width w. M' instances of parameterized Attention Pooling [27] are employed to extract global features. The pooled features are concatenated with the flattened feature map to obtain the global representations as

$$q^{\text{global}} = \text{Concat}\Big(\big\{\,\text{AttnPool}_i(\mathcal{F}_L)\big\}_{i=1}^{M'}\,,\ \text{Flatten}(\mathcal{F}_L)\Big). \quad (3)$$

ods [17, 47]；（2）非并行的自回归解码导致推理速度缓慢。例如，图像分类需要在给定图像条件下计算并比较所有类别名称的累积概率；（3）它们在多任务学习中还存在任务干扰问题，导致与任务特定模型相比性能下降。

或者，Uni-Perceivers [1, 48] 将不同任务表述为通过表示相似性为每个输入寻找最大似然目标，而不考虑其模态，从而能够同时支持生成任务和非生成任务。然而，它们仍然无法处理图像检测和分割任务。

3. 重新审视 Uni-Perceivers

感知任务的统一建模。Uni-Perceiver [1] 提出将不同任务重新表述为一个统一的最大似然估计问题。具体而言，每个任务都由一组输入和一组候选目标定义，这些输入和目标可以来自任意模态组合。输入和目标首先通过具有线性投影的模态特定分词器进行编码。随后，编码后的表示由对不同任务共享参数的模态无关解码器进行转换。给定一个输入，统一的任务目标被定义为在该输入条件下寻找具有最大似然的目标。

缓解任务干扰。采用完全共享参数的多任务学习可能会在不同任务之间引入干扰。Uni-Perceiver-MoE [48] 提出了条件 MoE 来解决任务干扰问题。具体而言，对于每个输入标记，都会根据特定的路由策略计算路由决策，从而稀疏地激活少量专家来处理该标记。输入标记的相应输出是由路由决策选中的这些专家的线性加权组合。条件 MoE 通过允许存在冲突的模态和任务使用独立参数，同时不引入任何任务特定模块，从而缓解了干扰问题。

局限性。尽管 Uni-Perceivers 旨在用统一架构处理不同任务，但由于其编码特征中缺乏定位信息，它无法处理检测和分割任务。同时，Uni-Perceivers 未集成现成的编码器模型，因此无法受益于现有的大规模预训练编码器。这可能会增加其对预训练数据和资源的需求，从而限制其性能。

4. 方法

4.1. 将图像编码为通用区域提议

大多数现有的通用模型 [1, 48] 将图像表示为固定大小的不重叠图像块。这种设计相当粗糙，在建模图像中不同大小和形状的物体方面存在局限，使其难以处理检测和分割等定位任务。

为了实现更具表达力且更灵活的定位建模，我们提出将输入图像编码为一系列通用区域候选。具体而言，给定一幅高度为 H、宽度为 W 的输入图像 $x \in \mathbb{R}^{H\times W}$，采用网络 $f_{\text{image}}(\cdot)$ 将该图像编码为全局表示和区域表示的拼接，如下所示：

$$f_{\text{image}}(x) = \text{Concat}\left(\{q_i^{\text{global}}\}_{i=1}^{M}, \{q_j^{\text{proposal}}\}_{j=1}^{N}\right), \quad (1)$$

其中，$q_i^{\text{global}} \in \mathbb{R}^d$ 是整幅图像的全局表示，$q_j^{\text{proposal}} \in \mathbb{R}^d$ 是图像中候选目标提议的区域表示。

遵循定位任务中的常见做法，首先采用图像骨干网络（例如 ResNet [12]）来提取多尺度特征图 $\{\mathcal{F}_l\}_{l=1}^{L}$，其中 L 是特征尺度的数量（例如，$L =$ 为 4）。

区域表示。在多尺度特征图 $\{\mathcal{F}_l\}_{l=1}^{L}$ 之上应用一个基于 Transformer [36] 的区域提议网络，以提取一组 O 个候选目标提议 $\{q_j^{\text{sem}}, q_j^{\text{box}}, q_j^{\text{mask}}\}_{j=1}^{O}$，其中 $q_j^{\text{sem}} \in \mathbb{R}^d$、$q_j^{\text{box}} \in \mathbb{R}^4$ 和 $q_j^{\text{mask}} \in \mathbb{R}^{H\times W}$ 分别是第 j 个提议的语义、边界框和分割掩码表示。该区域提议网络类似于 MaskDINO [17]，但仅考虑前景-背景二分类。详细实现见附录。然后将这三种表示融合为区域表示，如下所示：

$$q_j^{\text{proposal}} = q_j^{\text{sem}} + \mathcal{B}(q_j^{\text{box}}) + \mathcal{M}(q_j^{\text{mask}}), \quad (2)$$

其中 $\mathcal{B}$ 表示框坐标的位置编码。$\mathcal{M}$ 使用自适应平均池化将掩码预测缩放到 28×28 的大小。$\mathcal{B}$ 和 $\mathcal{M}$ 后面都接有线性投影，以匹配特征维度。

全局表示。全局表示是从最后尺度的特征图 $\mathcal{F}_L \in \mathbb{R}^{h\times w}$ 中提取的，其高度为 h、宽度为 w。采用 M' 个参数化注意力池化 [27] 实例来提取全局特征。将池化后的特征与展平的特征图拼接，以获得全局表示，如下所示：

$$q^{\text{global}} = \text{Concat}\Big(\big\{\,\text{AttnPool}_i(\mathcal{F}_L)\big\}_{i=1}^{M'}, \ \text{Flatten}(\mathcal{F}_L)\Big). \quad (3)$$

4.2. Encoding Text with Language Models

A Transformer [36]-based language model is used to encode textual data, such as category names in classification tasks, image descriptions in image-text retrieval tasks, and the vocabulary in image captioning tasks. Specifically, a BPE tokenizer [30] tokenizes the input text x into a sequence of word embeddings, and a Transformer encoder is employed to extracts the text feature sequence as

$$f_{\text{text}}(x) = \text{Concat}(q_1^{\text{text}}, q_2^{\text{text}}, \cdots, q_L^{\text{text}}) \tag{4}$$

where $q_i^{\text{text}} \in \mathbb{R}^d$ is the encoded feature of the i-th word, and L is the sequence length. In our implementation, we use a pre-trained RoBERTa $_{\text{BASE}}$ [20] as the text encoder, which is jointly tuned with the whole network.

4.3. General Task Adaptation

We follow Uni-Perceivers [1, 48] to formulate different tasks as a unified maximum likelihood estimation problem. Given an input $x \in \mathcal{X}$ and the candidate target set $\mathcal{Y}$, the task objective is defined as finding the target $\hat{y} \in \mathcal{Y}$ with the maximum likelihood as

$$\hat{y} = \arg\max_{y \in \mathcal{Y}} P(x, y), \tag{5}$$

where the likelihood $P(x, y)$ is estimated from the cosine similarity between the representations of x and y as

$$P(x, y) \propto \exp\left(\cos\Big(g \circ f(x)\ ,\ g \circ f(y)\Big)/\tau\right), \tag{6}$$

where $f(\cdot)$ is the modality-specific encoders f_{image} and f_{text} introduced in Sec. 4.1 and 4.2, respectively. $g(\cdot)$ is a modality-agnostic Transformer [36] network shared for different tasks, and $\tau > 0$ is a learnable temperature parameter.

Depending on task requirements, the modality-specific encoded representation for inputs x can be an image feature sequence $f_{\text{image}}(x)$, a text feature sequence $f_{\text{text}}(x)$, or their concatenation, with an additional <SPE> token inserted at the beginning. The encoded representation for targets y is constructed in the same way.

To obtain general task modeling capability, Uni-Perceiver v2 conducts multi-task learning on various uni-modal and multi-modal tasks. Denoting a set of K tasks as $\{\mathcal{X}_k, \mathcal{Y}_k\}_{k=1}^K$, where $\mathcal{X}_k$ and $\mathcal{Y}_k$ are the input set and target set of the k-th task, respectively. The training loss is

$$L = \sum_{k=1}^{K} s_k \mathop{\mathbb{E}}_{\{x,y\} \in \{\mathcal{X}_k, \mathcal{Y}_k\}} \left[-w_k \log \frac{P(x, y)}{\sum_{z \in \mathcal{Y}_k} P(x, z)} \right], \tag{7}$$

where s_k and w_k denote the sampling ratio and loss weight of the k-th task, respectively. The sampling ratio are normalized as $\sum_k s_k = 1$. We refer to Sec. 4.4 for detailed discussions of the sampling strategy. To mitigate the task interference in multi-task training, we follow Uni-Perceiver-MoE [48] to employ the Conditional MoEs with attribute-level routing strategy for effective multi-task training.

Tasks with Localization. Uni-Perceiver v2 can perform localization tasks such as object detection and instance segmentation by decoding the regional representations. Specifically, for each region proposal q_j^{proposal}, its outputted feature from the unified decoder $g(\cdot)$ will be compared with class embeddings to obtain the class prediction as in Eq. (5). The corresponding bounding box q_j^{box} and segmentation mask q_j^{mask} will serve as the localization predictions.

Tasks without Localization. Uni-Perceiver v2 can also handle tasks that do need localization predictions, *e.g.,* image classification, image captioning, image-text retrieval. It follows a similar formulation of Uni-Perceiver for these tasks with two major differences: (1) More expressive and flexible localization clues for images, better facilitating these tasks; (2) Both the image and text encoders can leverage off-the-shelf modality-specific pre-trained models, leading to better performance.

4.4. Sampling Strategy and Improved Optimization

Optimizing generalist models follows the paradigm of multi-task learning, which performs joint training on data from different tasks. Current methods usually mix all tasks in one training iteration [1, 23, 39]. Such *mixed sampling strategy* limits the batch-size of each task, which can be detrimental for tasks that benefit from large batch-size training (*e.g.,* image-text retrieval).

A straightforward solution is to sample only one task per iteration, which we refer as *unmixed sampling strategy*. It can achieve the largest training batch-size. However, when different iterations sample different tasks, the gradients would vary greatly due to the differences in data and tasks, which may bring potential instability to multi-task learning and performance deterioration.

To mitigate the instability issue of unmixed sampling strategy, we propose an improved optimizer for multi-task training, named as **MT-AdamW**. The core idea is to balance the gradient of each task, by normalizing the gradient of each iteration and compensating it according to the task sampling ratio.

Suppose the k-th task is sampled at timestep t, the vanilla AdamW [22] is modified to MT-AdamW by updating the parameters θ as follows:

$$\left\{\begin{array}{l} \mathbf{g}_t \leftarrow \nabla L_{t,k}\left(\theta_{t-1}\right) \\ \mathbf{m}_t{=}(1{-}\beta_1)\mathbf{m}_{t-1}{+}\beta_1\mathbf{g}_t \\ \mathbf{n}_t{=}(1{-}\beta_2)\mathbf{n}_{t-1}{+}\beta_2\mathbf{g}_t^2 \\ \theta_t{=}\theta_{t-1} - \alpha\frac{\mathbf{m}_t}{\sqrt{\mathbf{n}_t{+}\varepsilon}} \end{array}\right. \Rightarrow \left\{\begin{array}{l} \mathbf{g}_t \leftarrow \omega_k \frac{\nabla L_{t,k}\left(\theta_{t-1}\right)}{\|\nabla L_{t,k}\left(\theta_{t-1}\right)\|} \\ \mathbf{m}_t{=}(1{-}\beta_1)\mathbf{m}_{t-1}{+}\frac{\beta_1}{s_k}\mathbf{g}_t \\ \mathbf{n}_t{=}(1{-}\beta_2)\mathbf{n}_{t-1}{+}\frac{\beta_2}{s_k}\mathbf{g}_t^2 \\ \theta_t{=}\theta_{t-1} - \alpha\frac{\mathbf{m}_t}{\sqrt{\mathbf{n}_t{+}\varepsilon}} \end{array}\right.$$

4.2. 使用语言模型编码文本

使用基于 Transformer [36] 的语言模型对文本数据进行编码，例如分类任务中的类别名称、图文检索任务中的图像描述，以及图像字幕生成任务中的词汇表。具体而言，BPE 分词器 [30] 将输入文本 x 分词为一系列词嵌入，并采用 Transformer 编码器提取文本特征序列，如下所示：

$$f_{\text{text}}(x) = \text{Concat}(q_1^{\text{text}}, q_2^{\text{text}}, \cdots, q_L^{\text{text}}) \tag{4}$$

其中，$q_i^{\text{text}} \in \mathbb{R}^d$ 是第 i 个词的编码特征，L 是序列长度。在我们的实现中，我们使用预训练的 RoBERTa $_{\text{BASE}}$ [20] 作为文本编码器，并与整个网络联合微调。

4.3. 通用任务适配

我们遵循 Uni-Perceivers [1, 48]，将不同任务表述为统一的最大似然估计问题。给定输入 $x \in \mathcal{X}$ 和候选目标集 $\mathcal{Y}$，任务目标被定义为寻找具有最大似然的目标 $\hat{y} \in \mathcal{Y}$，即：

$$\hat{y} = \arg\max_{y \in \mathcal{Y}} P(x, y), \tag{5}$$

其中，似然 $P(x, y)$ 根据 x 和 y 的表示之间的余弦相似度估计为

$$P(x, y) \propto \exp\left(\cos\Big(g \circ f(x)\ ,\ g \circ f(y)\Big) / \tau \right), \tag{6}$$

其中，$f(\cdot)$ 分别是第 4.1 节和第 4.2 节中介绍的模态特定编码器 f_{image} 和 f_{text}。$g(\cdot)$ 是一个模态无关的 Transformer [36] 网络，在不同任务之间共享，而 $\tau > 0$ 是一个可学习的温度参数。

根据任务需求，输入 x 的模态特定编码表示可以是图像特征序列 $f_{\text{image}}(x)$、文本特征序列 $f_{\text{text}}(x)$，或它们的拼接，并在开头插入一个额外的 <SPE> 标记。目标 y 的编码表示以相同方式构建。

为了获得通用的任务建模能力，Uni-Perceiver v2 在各种单模态和多模态任务上进行多任务学习。将一组 K 个任务表示为 $\{\mathcal{X}_k, \mathcal{Y}_k\}_{k=1}^K$，其中 $\mathcal{X}_k$ 和 $\mathcal{Y}_k$ 分别是第 k 个任务的输入集和目标集。训练损失为

$$L = \sum_{k=1}^{K} s_k \mathop{\mathbb{E}}_{\{x,y\} \in \{\mathcal{X}_k, \mathcal{Y}_k\}} \left[-w_k \log \frac{P(x, y)}{\sum_{z \in \mathcal{Y}_k} P(x, z)} \right], \tag{7}$$

其中 s_k 和 w_k 分别表示第 k 个任务的采样比例和损失权重。采样比例被归一化为 $\sum_k s_k = 1$。详细内容请参见第 4.4 节。

采样策略的讨论。为了缓解多任务训练中的任务干扰，我们遵循 Uni-Perceiver-MoE [48]，采用具有属性级路由策略的条件 MoE，以实现有效的多任务训练。

带定位的任务。Uni-Perceiver v2 可以通过解码区域表示来执行目标检测和实例分割等定位任务。具体而言，对于每个区域提议 q_j^{proposal}，其由统一解码器 $g(\cdot)$ 输出的特征将与类别嵌入进行比较，以获得如式（5）所示的类别预测。相应的边界框 q_j^{box} 和分割掩码 q_j^{mask} 将作为定位预测。

无需定位的任务。Uni-Perceiver v2 也可以处理需要定位预测的任务，例如图像分类、图像字幕生成、图像-文本检索。对于这些任务，它遵循与 Uni-Perceiver 类似的表述方式，但有两个主要区别：（1）为图像提供更具表达力和灵活性的定位线索，从而更好地促进这些任务；（2）图像编码器和文本编码器都可以利用现成的模态特定预训练模型，从而获得更好的性能。

4.4. 采样策略与改进优化

优化通用模型遵循多任务学习范式，即在来自不同任务的数据上进行联合训练。当前方法通常在一次训练迭代中混合所有任务 [1, 23, 39]。这种混合采样策略限制了每个任务的批大小，这可能不利于那些受益于大批量训练的任务（例如，图文检索）。

一种直接的解决方案是每次迭代只采样一个任务，我们将其称为非混合采样策略。它可以实现最大的训练批大小。然而，当不同迭代采样不同任务时，由于数据和任务的差异，梯度会发生较大变化，这可能会给多任务学习带来潜在的不稳定性并导致性能下降。

为缓解非混合采样策略的不稳定性问题，我们提出了一种用于多任务训练的改进优化器，命名为 MT-AdamW。其核心思想是通过对每次迭代的梯度进行归一化，并根据任务采样比例对其进行补偿，从而平衡各个任务的梯度。

假设第 k 个任务在时间步 t 被采样，则将原始的 AdamW [22] 修改为 MT-AdamW，并按如下方式更新参数 θ：

$$\left\{\begin{array}{l} \mathbf{g}_t \leftarrow \nabla L_{t,k}\left(\theta_{t-1}\right) \\ \mathbf{m}_t = (1-\beta_1)\mathbf{m}_{t-1} + \beta_1 \mathbf{g}_t \\ \mathbf{n}_t = (1-\beta_2)\mathbf{n}_{t-1} + \beta_2 \mathbf{g}_t^2 \\ \theta_t = \theta_{t-1} - \alpha \frac{\mathbf{m}_t}{\sqrt{\mathbf{n}_t + \varepsilon}} \end{array}\right. \Rightarrow \left\{\begin{array}{l} \mathbf{g}_t \leftarrow \omega_k \frac{\nabla L_{t,k}\left(\theta_{t-1}\right)}{\|\nabla L_{t,k}\left(\theta_{t-1}\right)\|} \\ \mathbf{m}_t = (1-\beta_1)\mathbf{m}_{t-1} + \frac{\beta_1}{s_k} \mathbf{g}_t \\ \mathbf{n}_t = (1-\beta_2)\mathbf{n}_{t-1} + \frac{\beta_2}{s_k} \mathbf{g}_t^2 \\ \theta_t = \theta_{t-1} - \alpha \frac{\mathbf{m}_t}{\sqrt{\mathbf{n}_t + \varepsilon}} \end{array}\right.$$

where $L_{t,k}$ is the loss function for the sampled k-th task at timestep t, and α is the learning rate. The weight decay and bias corrections are omitted for simplicity. The original task gradients are first normalized to stabilize training. The scaling factor ω_k serves as the loss weight of the sampled task. Then the trimmed gradient $\mathbf{g}_t$ can be used to estimate the first moment $\mathbf{m}_t$ and second moment $\mathbf{n}_t$ of gradients in a moving average way. To further decouple the gradient contribution and sampling ratio s_k of each task, a task-specific compensation coefficient $1/s_k$ is used to unbias the estimation $\mathbf{m}_t$ and $\mathbf{n}_t$. In practice, if all tasks are expected to contribute equally, all scaling factors could be set as $\omega_k = 1$.

5. Experiments

5.1. Datasets

Uni-Perceiver v2 performs multi-task training on various tasks and public-available datasets to achieve the general task modeling capability. It uses similar datasets as in Uni-Perceiver [1]. Specifically, the image classification task is trained on ImageNet-1k [9] dataset. For objection detection and instance segmentation, COCO [19] is used for training. For image captioning and image-text retrieval, we use a combination of image-text-pair datasets: SBU Captions [25], Visual Genome [16], COCO Caption [8], CC3M [33], CC12M [5] and YFCC [14]. We also add the language modeling task during training, which is trained on BookCorpus [49] and English Wikipedia (Books&Wiki).

During the evaluation, we evaluate generalist models on the most representative datasets for the pillar vision and vision-language tasks listed in Tab. 1. Specifically, ImageNet-1k [9] and COCO Caption [8] are utilized to evaluate the performance of image classification and image caption, respectively. For image-text retrieval, COCO Caption and Flickr30k [26] are utilized. Note that Flickr30k is not involved in training. For objection detection and instance segmentation, COCO [19] is used to evaluate their performances. We put the licenses of all datasets in the Appendix.

5.2. Implementation Details

We implement three Uni-Perceiver v2 variants with different backbones, *i.e.,* ResNet-50 [12], Swin-Base [21], and Swin-Large. ResNet-50 is pre-trained on ImageNet-1k, and Swin-Base is pre-trained on ImageNet-21k. Swin-Large is firstly pre-trained on ImageNet-21k and then trained on the detection task with Object365 [32]. The number of feature scales L is set to 4 for all models. A Transformer [36]-based region proposal network is used to generate general region proposals, whose architecture and settings mainly follow Mask DINO [17]. However, we replace all multi-category classifiers with binary classifiers. In addition, the number of global attention pooling to extract global features is set to $M' = 10$. We choose the pre-trained RoBERTa $_{\text{BASE}}$ [20] as the text encoder, which is jointly tuned with the whole network. The unified decoder is also a Transformer-based network, whose parameters are initialized randomly and optimized from scratch. Its architecture follows the setting of the BERT $_{\text{BASE}}$ [10] model, but it only consists of 6 Transformer layers. To mitigate the task interference issue in multi-task learning, we also employ the attribute-level Conditional MoE [48] in all FFN layers of the unified decoder. Please refer to the Appendix for more details.

Unless specifically stated, we adopt the unmixed sampling strategy, which only samples one task for all GPUs per iteration. The MT-AdamW optimizer with a base learning rate of 0.0001 and a weight decay of 0.0001 is utilized. The learning rate of modality-specific encoders is multiplied by 0.1 since they have already been pre-trained. Uni-Perceiver v2 with Swin-Base and Swin-Large backbone is trained for 200,000 iterations on 32 and 64 NVIDIA A100 GPUs, respectively. The learning rate drops to $0.1\times$ at the 160,000 iterations. For models with ResNet-50, we only train them on 16 NVIDIA A100 GPUs for 150,000 iterations. For other training settings, please also refer to the Appendix.

5.3. Ablation Studies

In the following, we evaluate the key components of Uni-Perceiver v2 with ResNet-50 backbone by evaluating its performance on four tasks, *i.e.,* image detection on COCO, image classification on ImageNet-1k, image-text retrieval on COCO caption, and image captioning on COCO caption. The instance segmentation and language modeling tasks are not included to save training costs, and the YFCC dataset is also excluded from the training. Note that, the performance on these datasets are reported without any task-specific fine-tuning. If not stated, COCO detection pre-trained ResNet-50 is used for ablation studies to accelerate the convergence of multi-task training.

Effectiveness of Global and Regional Image Representations. Uni-Perceiver v2 encodes images as the concatenation of global and regional representations. To evaluate their effectiveness on different tasks, we conduct experiments that employ different representations, *i.e.,* only using global representations, only using regional representation only, and using both. Results in Tab. 2 show that: (1) regional representation is crucial for both captioning and retrieval tasks. We speculate that this is because regional proposals can provide localization clues, which is helpful to process both tasks. (2) Compared with regional-only representations, global representations deliver better results on the image classification task, which indicates global representations are important for image-level tasks. (3) Combining global and regional representation allows the two representations to complement each other, and thus achieve the best overall results on all tasks. Therefore, in our subsequent experiments, combining global and regional represen-

其中，$L_{t,k}$ 是在时间步 t 采样到的第 k 个任务的损失函数，α 是学习率。为简化起见，省略了权重衰减和偏置校正。首先对原始任务梯度进行归一化，以稳定训练。缩放因子 ω_k 作为采样任务的损失权重。然后，裁剪后的梯度 $\mathbf{g}_t$ 可用于以移动平均的方式估计梯度的一阶矩 $\mathbf{m}_t$ 和二阶矩 $\mathbf{n}_t$。为进一步解耦每个任务的梯度贡献和采样比例 s_k，使用任务特定的补偿系数 $1/s_k$ 对估计值 $\mathbf{m}_t$ 和 $\mathbf{n}_t$ 进行去偏。在实践中，如果期望所有任务贡献相同，则所有缩放因子都可以设置为 $\omega_k = 1$ 。

5. 实验

5.1. 数据集

Uni-Perceiver v2 在各种任务和公开可用数据集上进行多任务训练，以实现通用任务建模能力。它使用与 Uni-Perceiver [1] 类似的数据集。具体而言，图像分类任务在 ImageNet-1k [9] 数据集上训练。对于目标检测和实例分割，使用 COCO [19] 进行训练。对于图像字幕生成和图文检索，我们使用图文对数据集的组合：SBU Captions [25]、Visual Genome [16]、COCO Caption [8]、CC3M [33]、CC12M [5] 和 YFCC [14]。我们还在训练过程中加入语言建模任务，该任务在 BookCorpus [49] 和英文维基百科（Books&Wiki）上训练。

在评估过程中，我们在表 1 所列的支柱性视觉与视觉-语言任务中最具代表性的数据集上评估通用模型。具体而言，分别使用 ImageNet-1k [9] 和 COCO Caption [8] 来评估图像分类和图像描述生成的性能。对于图像-文本检索，使用 COCO Caption 和 Flickr30k [26]。需要注意的是，Flickr30k 未参与训练。对于目标检测和实例分割，使用 COCO [19] 来评估其性能。我们在附录中列出了所有数据集的许可证。

5.2. 实现细节

我们实现了三种采用不同骨干网络的 Uni-Perceiver v2 变体，即 ResNet-50 [12]、Swin-Base [21] 和 Swin-Large。ResNet-50 在 ImageNet-1k 上进行预训练，Swin-Base 在 ImageNet-21k 上进行预训练。Swin-Large 首先在 ImageNet-21k 上进行预训练，然后在 Object365 [32] 的检测任务上进行训练。所有模型的特征尺度数量 L 均设置为 4。我们使用基于 Transformer [36] 的区域提议网络来生成通用区域提议，其架构和设置主要遵循 Mask DINO [17]。然而，我们将所有多类别分类器替换为二分类器。此外，用于提取全局特征的全局注意力池化数量设置为 $M' = 10$。我们选择预训练的 RoBERTa $_{\text{BASE}}$ [20] 作为文本编码器，它与整个网络共同调优。统一解码器也是一个基于 Transformer 的网络，其参数随机初始化并从头开始优化。其架构遵循 BERT $_{\text{BASE}}$ [10] 模型的设置，但仅包含 6 个 Transformer 层。为缓解多任务学习中的任务干扰问题，我们还在统一解码器的所有 FFN 层中采用了属性级条件 MoE [48]。更多细节请参见附录。

除非特别说明，我们采用非混合采样策略，即每次迭代仅为所有 GPU 采样一个任务。使用 MT-AdamW 优化器，基础学习率为 0.0001，权重衰减为 0.0001。由于模态特定编码器已经过预训练，其学习率乘以 0.1 。采用 Swin-Base 和 Swin-Large 骨干网络的 Uni-Perceiver v2 分别在 32 块和 64 块 NVIDIA A100 GPU 上训练 200,000 次迭代。学习率在第 160,000 次迭代时下降到 0.1×。对于使用 ResNet-50 的模型，我们仅在 16 块 NVIDIA A100 GPU 上训练 150,000 次迭代。其他训练设置也请参见附录。

5.3. 消融实验

在下文中，我们通过评估采用 ResNet-50 骨干网络的 Uni-Perceiver v2 在四项任务上的性能来评估其关键组件，即 COCO 上的图像检测、ImageNet-1k 上的图像分类、COCO caption 上的图文检索以及 COCO caption 上的图像描述生成。为节省训练成本，未包含实例分割和语言建模任务，并且训练中也排除了 YFCC 数据集。需要注意的是，这些数据集上的性能是在没有任何特定任务微调的情况下报告的。若未特别说明，消融研究使用经过 COCO 检测预训练的 ResNet-50，以加速多任务训练的收敛。

全局与区域图像表征的有效性。Uni-Perceiver v2 将图像编码为全局表征与区域表征的拼接。为了评估它们在不同任务上的有效性，我们进行了采用不同表征的实验，即仅使用全局表征、仅使用区域表征，以及同时使用二者。表 2 中的结果表明：（1）区域表征对于图像描述生成和检索任务都至关重要。我们推测，这是因为区域候选框能够提供定位线索，有助于处理这两类任务。（2）与仅使用区域表征相比，全局表征在图像分类任务上取得了更好的结果，这表明全局表征对于图像级任务非常重要。（3）结合全局表征与区域表征能够使两种表征相互补充，从而在所有任务上取得最佳的整体结果。因此，在我们后续的实验中，结合全局和区域表征

Representation Types	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Global	-	76.8	46.3	34.6	28.8
Regional	48.2	75.9	**52.3**	**39.2**	**31.2**
Global + Regional	**49.9**	**76.9**	51.3	38.8	30.6

Table 2. Ablation of different representation types for general region proposals. Results are reported on object detection (mAP), image classification (Acc), image-text retrieval (I2T R@1 and T2I R@1), and image caption (BLEU-4).

Tasks	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Single Task	50.1	76.1	50.0	37.6	30.2
All Tasks	49.8	76.3	46.0	34.7	28.9
w/o Detection	-	76.6 (+0.3)	47.0 (+1.0)	34.6 (−0.1)	30.4 (+0.5)
w/o Classification	50.1 (+0.3)	-	51.6 (+5.6)	38.6 (+3.9)	25.9 (−3.0)
w/o Retrieval	49.5 (−0.3)	76.3 (+0.0)	-	-	27.4 (−1.5)
w/o Captioning	49.7 (−0.1)	76.3 (+0.0)	51.2 (+5.2)	38.3 (+3.6)	-
All Tasks w/ MoE	49.9 (+0.1)	76.9 (+0.6)	51.3 (+5.3)	38.8 (+4.1)	30.6 (+0.7)

Table 3. Ablation of collaboration and interference between tasks. All experiments except for the last line do not employ Conditional MoEs. In the brackets are the gaps to the "All Tasks" counterpart. In green and red are the gaps of at least ±0.5 point.

tations is taken as the default setting.

Task Collaboration and Interference. To analyze the collaboration and interference between different tasks, we conduct experiments by removing each task independently from the joint-training tasks in Tab. 3. If the removal of one task can improve (or degrade) the performance of another task, it can reflect that the former task is detrimental (or beneficial) to the latter one during joint training. For a fair comparison, the Conditional MoEs are not employed except for the last experiment. Results show that without MoEs, other tasks have negative impacts on the training of image-text retrieval. However, the image-text retrieval task could promote the performance of image captioning. The image classification task is also very helpful to image captioning, yet the reverse has no obvious effect. It should be noted that all models employ an image encoder pre-trained on COCO detection, thereby all these tasks can benefit from the pre-trained region proposal network. The results indicate that task interference indeed exists in the multi-task training of generalist models and is more common than task collaboration, suggesting the importance of addressing the task interference issue. By employing Conditional MoEs, the task interference is largely mitigated, resulting in improved results on all tasks.

Sampling Strategy and Improved Optimization. We evaluate the effectiveness of the unmixed sampling strategy (*i.e.,* sampleing one task for each iteration) and the proposed MT-AdamW optimizer in Tab. 4. From the results,

Task Sampling	Gather Feature	MT-AdamW Optimizer	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
mixed			49.6	76.7	40.1	31.9	27.6
unmixed			49.2	76.6	39.8	30.9	27.5
unmixed	✓		49.3	76.8	50.4	37.3	27.6
unmixed	✓	✓	**49.9**	**76.9**	**51.3**	**38.8**	**30.6**

Table 4. Ablation of sampling strategies and improved optimizer. "mixed" means mixing different tasks' data in one iteration, while "unmixed" denotes that only one task's data is sampled in one iteration. "Gather Feature" means that negative samples for retrieval tasks are collected synchronously across GPUs.

Pretrained Method	Pretrained Data	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Supervised	IN-1k	45.7	76.8	51.2	38.9	27.3
Supervised	IN-21k	48.3	**80.1**	55.1	41.2	30.2
Supervised	IN-1k & COCO	**49.9**	76.9	51.3	38.8	30.6
MoCo v2	IN-1k	48.3	75.0	54.8	40.5	29.6
CLIP	CLIP data	47.2	73.8	**55.3**	**41.3**	**32.0**

Table 5. Ablation of different pre-trained image encoders.

we observe that the vanilla unmixed sampling strategy that computing the contrastive loss with samples on each GPU have slightly adverse effect on the learning of all tasks when compared with the mixed sampling strategy. With the batch size increased by gathering features across all GPUs, the performance of retrieval tasks can be largely improved. Further introducing the MT-AdamW optimizer leads to more stable multi-task training and consistently improved performance across all tasks.

Effects of Different Image Encoder Pre-training. By integrating off-the-shelf encoder models, Uni-Perceiver v2 is capable of leveraging existing large-scale pre-trained encoders. To analyze the effects of different pre-training, we employ different pre-trained models for image encoders. For models with supervised pre-training, we employ ResNet-50 pre-trained on ImageNet-1k, on ImageNet-21k, or consecutively pre-trained on ImageNet-1k and COCO. For models with weakly-supervised or unsupervised pre-training, we employ ResNet-50 pre-trained with MoCo v2 [7] or CLIP [27]. Tab. 5 demonstrates that different pre-training data and methods of image encoders benefit different downstream tasks. Specifically, supervised pre-training methods show the most obvious benefits on downstream tasks similar to it, *e.g.,* ImageNet-21k pre-training delivers the best results on ImageNet-1k classification. Besides, the pre-training on large-scale supervised (ImageNet-21k), weakly-supervised or unsupervised data (CLIP and MoCo v2) is more helpful to vision-language tasks such image-text retrieval and image captioning, which possibly thanks to more general representations.

Representation Types	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Global	-	76.8	46.3	34.6	28.8
Regional	48.2	75.9	**52.3**	**39.2**	**31.2**
Global + Regional	**49.9**	**76.9**	51.3	38.8	30.6

表 2. 用于通用区域提议的不同表示类型的消融实验。结果报告于目标检测（mAP）、图像分类（Acc）、图像-文本检索（I2T R@1 和 T2I R@1）以及图像描述（BLEU-4）。

Tasks	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Single Task	50.1	76.1	50.0	37.6	30.2
All Tasks	49.8	76.3	46.0	34.7	28.9
w/o Detection	-	76.6 (+0.3)	47.0 (+1.0)	34.6 (−0.1)	30.4 (+0.5)
w/o Classification	50.1 (+0.3)	-	51.6 (+5.6)	38.6 (+3.9)	25.9 (−3.0)
w/o Retrieval	49.5 (−0.3)	76.3 (+0.0)	-	-	27.4 (−1.5)
w/o Captioning	49.7 (−0.1)	76.3 (+0.0)	51.2 (+5.2)	38.3 (+3.6)	-
All Tasks w/ MoE	49.9 (+0.1)	76.9 (+0.6)	51.3 (+5.3)	38.8 (+4.1)	30.6 (+0.7)

表 3. 任务之间协作与干扰的消融实验。除最后一行外，所有实验均未采用条件式 MoE。括号中为与“All Tasks”对应项的差距。绿色和红色表示至少 ±0.5 分的差距。

tations 被作为默认设置。

任务协作与干扰。为分析不同任务之间的协作与干扰，我们通过从表 3 的联合训练任务中分别独立移除每个任务来进行实验。如果移除某一任务能够提升（或降低）另一任务的性能，则可以反映出在联合训练过程中，前一任务对后一任务是有害的（或有益的）。为公平比较，除最后一个实验外，均不采用 Conditional MoEs。结果表明，在没有 MoEs 的情况下，其他任务会对图文检索的训练产生负面影响。然而，图文检索任务能够提升图像描述生成的性能。图像分类任务对图像描述生成也非常有帮助，但反过来并无明显效果。需要注意的是，所有模型都采用了在 COCO 检测上预训练的图像编码器，因此所有这些任务都可以受益于预训练的区域提议网络。结果表明，任务干扰确实存在于通用模型的多任务训练中，并且比任务协作更为常见，这说明解决任务干扰问题十分重要。通过采用 Conditional MoEs，任务干扰在很大程度上得到缓解，从而在所有任务上都取得了更好的结果。

采样策略与改进优化。我们在表 4 中评估了非混合采样策略（即每次迭代采样一个任务）以及所提出的 MT-AdamW 优化器的有效性。从结果来看，

Task Sampling	Gather Feature	MT-AdamW Optimizer	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
mixed			49.6	76.7	40.1	31.9	27.6
unmixed			49.2	76.6	39.8	30.9	27.5
unmixed	✓		49.3	76.8	50.4	37.3	27.6
unmixed	✓	✓	**49.9**	**76.9**	**51.3**	**38.8**	**30.6**

表4. 采样策略和改进优化器的消融实验。“mixed”表示在一次迭代中混合不同任务的数据，而“unmixed”表示在一次迭代中只采样一个任务的数据。“Gather Feature”表示检索任务的负样本会在多个 GPU 间同步收集。

Pretrained Method	Pretrained Data	COCO Detection	ImageNet-1k Classification	COCO Retrieval		COCO Caption
Supervised	IN-1k	45.7	76.8	51.2	38.9	27.3
Supervised	IN-21k	48.3	**80.1**	55.1	41.2	30.2
Supervised	IN-1k & COCO	**49.9**	76.9	51.3	38.8	30.6
MoCo v2	IN-1k	48.3	75.0	54.8	40.5	29.6
CLIP	CLIP data	47.2	73.8	**55.3**	**41.3**	**32.0**

表5. 不同预训练图像编码器的消融实验。

我们观察到，与混合采样策略相比，普通的未混合采样策略（即使用每个 GPU 上的样本计算对比损失）会对所有任务的学习产生轻微的不利影响。通过在所有 GPU 间聚合特征来增大批量大小，检索任务的性能可以得到大幅提升。进一步引入 MT-AdamW 优化器后，多任务训练更加稳定，并且在所有任务上都能持续提升性能。

不同图像编码器预训练的影响。通过集成现成的编码器模型，Uni-Perceiver v2 能够利用已有的大规模预训练编码器。为分析不同预训练的影响，我们为图像编码器采用了不同的预训练模型。对于采用监督预训练的模型，我们使用在 ImageNet-1k 上预训练的 ResNet-50、在 ImageNet-21k 上预训练的 ResNet-50，或依次在 ImageNet-1k 和 COCO 上预训练的 ResNet-50。对于采用弱监督或无监督预训练的模型，我们使用通过 MoCo v2 [7] 或 CLIP [27] 预训练的 ResNet-50。表 5 表明，图像编码器不同的预训练数据和方法会有利于不同的下游任务。具体而言，监督预训练方法在与其相似的下游任务上表现出最明显的收益，例如，ImageNet-21k 预训练在 ImageNet-1k 分类上取得了最佳结果。此外，在大规模监督数据（ImageNet-21k）、弱监督或无监督数据（CLIP 和 MoCo v2）上的预训练，对图文检索和图像描述生成等视觉-语言任务更有帮助，这可能得益于更通用的表征。

Methods	#params	Image Classification	Object Detection	Instance Segmentation	Image Captioning		Text Retrieval		Image Retrieval	
		ImageNet-1k Acc	COCO mAP	COCO mAP	COCO B@4	COCO CIDEr	COCO R@1	Flickr30k R@1	COCO R@1	Flickr30k R@1
Pix2Seq v2 [6]	132M	-	46.5	38.2	34.9	-	-	-	-	-
UniTab [43]	185M	-	-	-	-	115.8	-	-	-	-
Unified-IO $_{\text{LARGE}}$ [23]	776M	71.8	-	-	-	-	-	-	-	-
Unified-IO $_{\text{XL}}$ [23]	2.9B	79.1	-	-		122.3	-	-	-	-
Flamingo-3B [2]	3.2B	-	-	-	-	-	65.9	89.3	48.0	79.5
Uni-Perceiver $_{\text{BASE}}$ [1]	124M	79.2	-	-	32.0	-	64.9	82.3	50.7	71.1
Uni-Perceiver $_{\text{LARGE}}$ [1]	354M	82.7	-	-	35.3	-	67.8	83.7	54.1	74.2
Uni-Perceiver-MoE $_{\text{BASE}}$ [48]	167M	80.3	-	-	33.2	-	64.6	82.1	51.6	72.4
Uni-Perceiver-MoE $_{\text{LARGE}}$ [48]	505M	83.4	-	-	35.5	-	67.9	83.6	55.3	75.9
Uni-Perceiver-v2 $_{\text{BASE}}$	308M	86.3	58.6	50.6	35.4	116.9	71.8	88.1	55.6	73.8
Uni-Perceiver-v2 $_{\text{LARGE}}$	446M	**87.2** (+3.8)	**61.9** (+15.4)	**53.6** (+15.4)	**36.5** (+1.6)	**122.5** (+0.2)	**75.0** (+7.1)	**89.3** (+0.0)	**58.5** (+3.2)	**79.6** (+0.1)

Table 6. Comparison of our Uni-Perceiver v2 to recent generalist models on six pillar visual and visual-linguistic tasks listed in Tab. 1. Note that we only report the results without any task-specific fine-tuning. Uni-Perceiver v2 is the the first generalist model to support all these pillar tasks and can achieve competitive results without any task-specific adaption. Some generalist models that only report results with task-specific fine-tuning are not included, *e.g.,* , OFA [39] and GIT [38]. "#params" is the number of parameters required during model deployment for cross-modal tasks. Results with the best performance are in **bold**, and previous SoTA results are underlined.

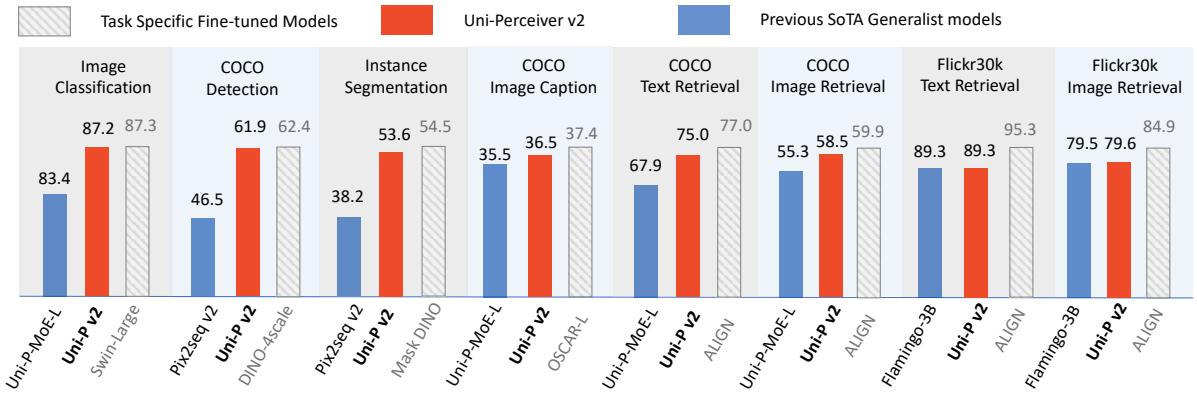


Figure 2. Comparison with generalist models and commonly-recognized strong task-specific models on pillar vision and vision-language tasks. For generalist models including Uni-Perceiver v2, we only report the results without any task-specific fine-tuning. Uni-Perceiver v2 (Uni-P v2) is compared with competitive specialized models, *i.e.,* Swin-large [21], DINO [47], Mask DINO [17], OSCAR-L [18] and ALIGN [13], and previous SoTA generalists, *i.e.,* Uni-P-MoE-L [48], Pix2seq v2 [6], and Flamingo-3B [2].

5.4. Main Results

To further verify the effectiveness of Uni-Perceiver v2, we incorporate more powerful backbones including Swin-Base and Swin-Large, denoted as Uni-Perceiver-v2 $_{\text{BASE}}$ and Uni-Perceiver-v2 $_{\text{LARGE}}$, respectively. In addition to the tasks included in the ablation studies, we also incorporate instance segmentation on COCO, language modeling on Books&Wiki, and image captioning / image-text retrieval on YFCC for larger-scale multi-task training.

Comparison with existing Generalist Models. We list the performance of Uni-Perceiver v2 and other generalist models on pillar vision and vision-language tasks in Tab. 6. Since generalist models aim to process different tasks with shared architecture and parameters, the task-specific fine-tuning will lose the general modeling ability. We report the performance of the shared models without any task-specific adaptation. Specifically, Uni-Perceiver-v2 $_{\text{BASE}}$ can outperform all previous generalist models on all tasks except the Flickr30k retrieval, even if some methods have $> 10\times$ model parameters, *e.g.,* Unified-IO$_{\text{XL}}$ and Flamingo-3B. The performance disadvantage on Flicker30k may be due to the use of private data by Flamingo-3B. Further Scaling up to Swin-Large backbone, Uni-Perceiver-v2 $_{\text{LARGE}}$ obtains the best performance on all tasks. Thanks to the flexibility of general region proposals, Uni-Perceiver v2 supports most pillar tasks among generalist models and can achieve competitive results consistently, which indicates the supe-

Methods	#params	Image Classification	Object Detection	Instance Segmentation	Image Captioning		Text Retrieval		Image Retrieval	
		ImageNet-1k Acc	COCO mAP	COCO mAP	COCO B@4	COCO CIDEr	COCO R@1	Flickr30k R@1	COCO R@1	Flickr30k R@1
Pix2Seq v2 [6]	132M	-	46.5	38.2	34.9	-	-	-	-	-
UniTab [43]	185M	-	-	-	-	115.8	-	-	-	-
Unified-IO $_{\text{LARGE}}$ [23]	776M	71.8	-	-	-	-	-	-	-	-
Unified-IO $_{\text{XL}}$ [23]	2.9B	79.1	-	-		122.3	-	-	-	-
Flamingo-3B [2]	3.2B	-	-	-	-	-	65.9	89.3	48.0	79.5
Uni-Perceiver $_{\text{BASE}}$ [1]	124M	79.2	-	-	32.0	-	64.9	82.3	50.7	71.1
Uni-Perceiver $_{\text{LARGE}}$ [1]	354M	82.7	-	-	35.3	-	67.8	83.7	54.1	74.2
Uni-Perceiver-MoE $_{\text{BASE}}$ [48]	167M	80.3	-	-	33.2	-	64.6	82.1	51.6	72.4
Uni-Perceiver-MoE $_{\text{LARGE}}$ [48]	505M	83.4	-	-	35.5	-	67.9	83.6	55.3	75.9
Uni-Perceiver-v2 $_{\text{BASE}}$	308M	86.3	58.6	50.6	35.4	116.9	71.8	88.1	55.6	73.8
Uni-Perceiver-v2 $_{\text{LARGE}}$	446M	**87.2** (+3.8)	**61.9** (+15.4)	**53.6** (+15.4)	**36.5** (+1.6)	**122.5** (+0.2)	**75.0** (+7.1)	**89.3** (+0.0)	**58.5** (+3.2)	**79.6** (+0.1)

表 6. 我们的 Uni-Perceiver v2 与近期通用模型在表 1 所列六项核心视觉和视觉-语言任务上的比较。请注意，我们仅报告未进行任何任务特定微调的结果。Uni-Perceiver v2 是首个支持所有这些核心任务的通用模型，并且无需任何任务特定适配即可取得有竞争力的结果。一些仅报告经过任务特定微调结果的通用模型未被纳入，例如 OFA [39] 和 GIT [38]。“#params”表示跨模态任务中模型部署所需的参数数量。最佳性能结果以粗体标出，先前的 SoTA 结果以下划线标出。

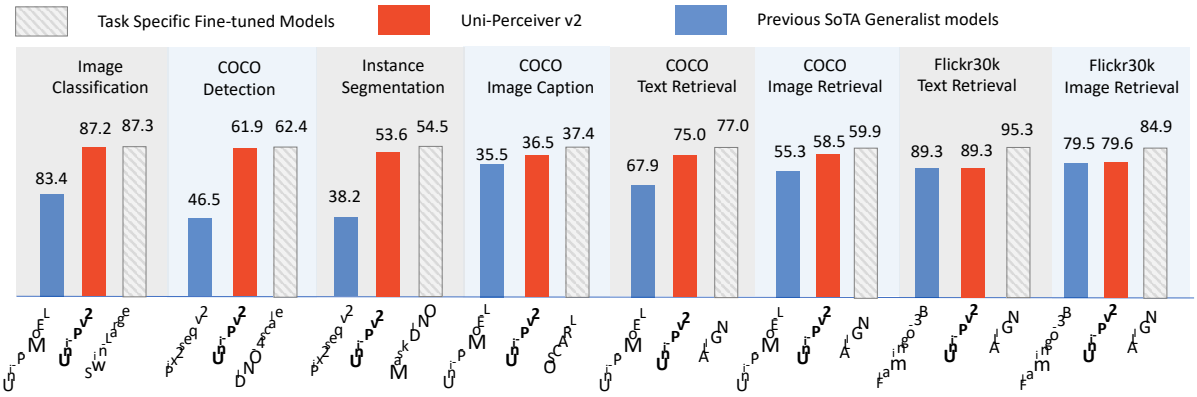


图 2. 在核心视觉和视觉-语言任务上，与通用模型以及公认的强任务专用模型进行比较。对于包括 Uni-Perceiver v2 在内的通用模型，我们仅报告未进行任何任务专用微调的结果。将 Uni-Perceiver v2（Uni-P v2）与具有竞争力的专用模型进行比较，即 Swin-large [21]、DINO [47]、Mask DINO [17]、OSCAR-L [18] 和 ALIGN [13]，以及先前的 SoTA 通用模型，即 Uni-P-MoE-L [48]、Pix2seq v2 [6] 和 Flamingo-3B [2]。

5.4. 主要结果

为进一步验证 Uni-Perceiver v2 的有效性，我们引入了更强大的骨干网络，包括 Swin-Base 和 Swin-Large，分别记为 Uni-Perceiver-v2 $_{\text{BASE}}$ 和 Uni-Perceiver-v2 $_{\text{LARGE}}$。除消融研究中包含的任务外，我们还纳入了 COCO 上的实例分割、Books&Wiki 上的语言建模，以及 YFCC 上的图像描述生成/图文检索，用于更大规模的多任务训练。

与现有通用模型的比较。我们在表 6 中列出了 Uni-Perceiver v2 以及其他通用模型在核心视觉和视觉-语言任务上的性能。由于通用模型旨在处理不同任务，并且共享架构和参数后，针对特定任务的微调会丧失通用建模能力。我们报告了共享模型在未进行任何特定任务适配情况下的性能。具体而言，Uni-Perceiver-v2 $_{\text{BASE}}$ 在除 Flickr30k 检索之外的所有任务上都能优于以往所有通用模型，即使某些方法拥有 $> 10\times$ 的模型参数，例如 Unified-IO$_{\text{XL}}$ 和 Flamingo-3B。在 Flicker30k 上的性能劣势可能是由于 Flamingo-3B 使用了私有数据。进一步扩展到 Swin-Large 骨干网络后，Uni-Perceiver-v2 $_{\text{LARGE}}$ 在所有任务上都取得了最佳性能。得益于通用区域提议的灵活性，Uni-Perceiver v2 在通用模型中支持大多数核心任务，并且能够持续取得有竞争力的结果，这表明其优越-

rior general modeling performance of Uni-Perceiver v2 in both versatility and performance.

Comparison with Specialized Models. We compare Uni-Perceiver v2 with commonly-recognized strong baseline models and previous SoTA generalist models on the pillar tasks in Tab. 2. The results show that Uni-Perceiver v2 significantly decreases the performance gap between generalist models and commonly-recognized strong baselines, which need task-specific fine-tuning. It can achieve comparable results across all tasks except the retrieval task on Flickr30K, which we suspect is because ALIGN [13] use 1.8B private image-text pairs, which is much larger than our training data. In contrast, Uni-Perceiver v2 uses only public data for training.

6. Conclusion

We propose Uni-Perceiver v2, which is the first generalist model that achieves competitive results on major large-scale vision and vision-language tasks. After being jointly trained on single-modal and multi-modal tasks, Uni-Perceiver v2 achieves competitive performance on a broad range of downstream tasks. As for **limitations**, our method has not been verified on image generation tasks due to limited computational resources.

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Uni-Perceiver v2 在通用性和性能方面都展现出更优越的通用建模性能。

与专用模型的比较。我们在表 2 的支柱任务上，将 Uni-Perceiver v2 与公认的强基线模型以及以往的 SoTA 通用模型进行了比较。结果表明，Uni-Perceiver v2 显著缩小了通用模型与公认强基线之间的性能差距，而这些强基线需要针对具体任务进行微调。除 Flickr30K 上的检索任务外，它在所有任务上都能取得相当的结果；我们推测这是因为 ALIGN [13] 使用了 18 亿个私有图文对，其规模远大于我们的训练数据。相比之下，Uni-Perceiver v2 仅使用公开数据进行训练。

6. 结论

我们提出了 Uni-Perceiver v2，这是首个在主要大规模视觉和视觉-语言任务上取得具有竞争力结果的通用模型。在单模态和多模态任务上进行联合训练后，Uni-Perceiver v2 在广泛的下游任务上取得了具有竞争力的性能。至于局限性，由于计算资源有限，我们的方法尚未在图像生成任务上得到验证。

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A. Architecture Details of the Image Encoder

As shown in Fig. 3, our Uni-Perceiver v2 consists of three main parts: the image encoder, the text encoder, and the unified decoder. In this section, we describe the architecture details of the image encoder.

Backbone Network. Given an input image $x \in \mathbb{R}^{H\times W}$ with height H and width W, a backbone network (*e.g.,* ResNet [12], Swin-Transformer [21]) is firstly employed to extract the multi-scale feature maps $\{\mathcal{F}_l\}_{l=0}^{L-1}$, where $L = 4$ is the number of features scales, and the spatial shapes of the feature maps are $\frac{H}{4}\times\frac{W}{4}$, $\frac{H}{8}\times\frac{W}{8}$, $\frac{H}{16}\times\frac{W}{16}$, and $\frac{H}{32}\times\frac{W}{32}$. The feature maps are transformed by 1×1 convolutions to match the hidden dimension of the following Transformer-based region proposal network. The transformed feature maps are denoted as $\mathcal{F}'_l$. An additional 3×3 stride 2 convolution layer is applied on $\mathcal{F}_3$ to extract a smaller feature map $\mathcal{F}'_4 \in \mathbb{R}^{\frac{H}{64}\times\frac{W}{64}\times d}$. $d = 256$ is the hidden dimension of the Transformer.

Region Proposal Network. A Transformer-based region proposal network is applied on top of the multi-scale feature maps to generate regional representations. Specifically, in the 4-scale setting which is adopted by Uni-Perceiver v2, the input of the Transformer encoder is the backbone feature maps except the first scale $\{\mathcal{F}'_l\}_{l=1}^{L=4}$. A deformable Transformer [59] encoder is employed to extract multi-scale encoded features $\{\mathcal{F}_l^{\text{enc}}\}_{l=1}^{L=4}$ whose spatial shapes and dimensions are the same as the corresponding input features. To generate the region proposals, we apply a deformalbe Transformer decoder on the multi-scale encoded features. To construct the N input object queries of the Transformer decoder (*e.g.,* $N = 900$), we predict the objectness and bounding boxes of each feature pixel in the encoded feature maps $\{\mathcal{F}_l^{\text{enc}}\}_{l=1}^{L=4}$, and select top-$N$ features based on their objectness. The selected features are added to N randomly initialized object queries as the input of the Transformer decoder, and their locations serve as the initial guess of the bounding boxes of the region proposals.

The Transformer decoder generates a set of N candidate object proposals $\{q_j^{\text{sem}}, q_j^{\text{box}}, q_j^{\text{mask}}\}_{j=1}^{N}$, where $q_j^{\text{sem}} \in \mathbb{R}^d$, $q_j^{\text{box}} \in \mathbb{R}^4$, and $q_j^{\text{mask}} \in \mathbb{R}^{H\times W}$ are the semantic, bounding box, and segmentation mask representations of the j-th proposal, respectively. Following Mask2Former [51] and MaskDINO [17], the segmentation mask representations are obtained by the dot product of the final-layer hidden

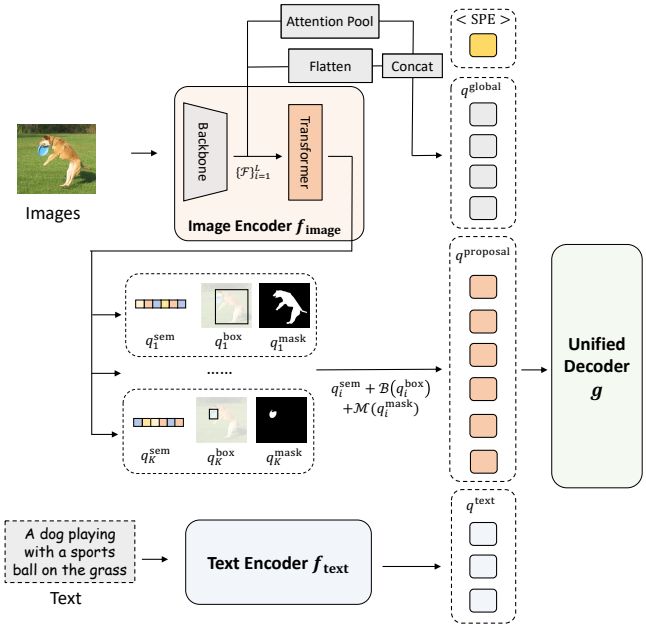


Figure 3. Architecture overview of our Uni-Perceiver v2.

state of the j-th proposal q_j and a per-pixel feature map,

$$q_i^{\text{mask}} = \text{Upsample}\Big(\text{MLP}(q_i)\odot\mathcal{R}\big(\mathcal{G}(\mathcal{F}_0)+\mathcal{H}(\mathcal{F}_1^{\text{enc}})\big)\Big), \tag{8}$$

where $\mathcal{G}$ is a 1×1 convolution layer followed by a Group Normalization (GN) [58], $\mathcal{H}$ is a 1×1 convolution followed by a GN and a bilinear upsampling, and $\mathcal{R}$ is a 3×3 convolution followed by a GN, a ReLU, and a 1×1 convolution.

The regional representations are obtained by fusing the semantic, bounding box, and segmentation mask representations,

$$q_j^{\text{proposal}} = q_j^{\text{sem}} + \mathcal{B}(q_j^{\text{box}}) + \mathcal{M}(q_j^{\text{mask}}), \tag{9}$$

where $\mathcal{B}$ denotes the positional encoding of box coordinates. $\mathcal{M}$ uses adaptive average pooling to scale the mask predictions to the size of 28×28. Both $\mathcal{B}$ and $\mathcal{M}$ are followed by linear projections to match the feature dimension. Note that the bounding box and segmentation mask representations are detached before fusing.

To reduce the computational cost, we predict objectness for each proposal q_j^{proposal}, and select the top-O proposals as the final regional representations. O is set as 200 by default in Uni-Perceiver v2.

Loss Function. In non-localization tasks such as image classification, the supervision is applied only on the final predictions of the unified decoder as Eq. 7, and there is no special supervision for the proposal generation of the image encoder. In localization tasks such as object detection, additional supervisions are applied for the training of the region proposal network. Specifically, we adopt the con-

ditional moes. *arXiv preprint arXiv:2206.04674*, 2022. 2, 4, 5, 6, 8, 12, 13

[49] Yukun Zhu, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. Aligning books and movies: Towards story-like visual explanations by watching movies and reading books. In *ICCV*, pages 19–27, 2015. 6, 12, 13

A. 图像编码器的架构细节

如图 3 所示，我们的 Uni-Perceiver v2 由三个主要部分组成：图像编码器、文本编码器和统一解码器。在本节中，我们描述图像编码器的架构细节。

骨干网络。给定一个高度为 H、宽度为 W 的输入图像 $x \in \mathbb{R}^{H\times W}$，首先采用骨干网络（例如 ResNet [12]、Swin-Transformer [21]）来提取多尺度特征图 $\{\mathcal{F}_l\}_{l=0}^{L-1}$，其中 $L=4$ 是特征尺度的数量，特征图的空间形状分别为 $\frac{H}{4}\times\frac{W}{4}$、$\frac{H}{8}\times\frac{W}{8}$、$\frac{H}{16}\times\frac{W}{16}$ 和 $\frac{H}{32}\times\frac{W}{32}$。通过 1×1 卷积对特征图进行变换，以匹配后续基于 Transformer 的区域提议网络的隐藏维度。变换后的特征图记为 $\mathcal{F}'_l$。在 $\mathcal{F}_3$ 上应用一个额外的 3×3、步幅为 2 的卷积层，以提取更小的特征图 $\mathcal{F}'_4 \in \mathbb{R}^{\frac{H}{64}\times\frac{W}{64}\times d}$。$d=256$ 是 Transformer 的隐藏维度。

区域候选网络。在多尺度特征图之上应用基于 Transformer 的区域候选网络，以生成区域表示。具体而言，在 Uni-Perceiver v2 采用的 4 尺度设置中，Transformer 编码器的输入是除第一尺度 $\{\mathcal{F}'_l\}_{l=1}^{L=4}$ 之外的骨干网络特征图。采用可变形 Transformer [59] 编码器来提取多尺度编码特征 $\{\mathcal{F}_l^{\text{enc}}\}_{l=1}^{L=4}$，其空间形状和维度与相应的输入特征相同。为了生成区域候选，我们在多尺度编码特征上应用可变形 Transformer 解码器。为了构建 Transformer 解码器的 N 个输入对象查询（例如，$N=900$），我们预测编码特征图 $\{\mathcal{F}_l^{\text{enc}}\}_{l=1}^{L=4}$ 中每个特征像素的目标性和边界框，并根据其目标性选择前 N 个特征。所选特征被添加到 N 个随机初始化的对象查询中，作为 Transformer 解码器的输入，并且它们的位置作为区域候选边界框的初始估计。

Transformer 解码器生成一组 N 个候选目标提案 $\{q_j^{\text{sem}}, q_j^{\text{box}}, q_j^{\text{mask}}\}_{j=1}^{N}$，其中 $q_j^{\text{sem}} \in \mathbb{R}^d$、$q_j^{\text{box}} \in \mathbb{R}^4$ 和 $q_j^{\text{mask}} \in \mathbb{R}^{H\times W}$ 分别是第 j 个提案的语义、边界框和分割掩码表示。遵循 Mask2Former [51] 和 MaskDINO [17]，分割掩码表示通过最终层隐藏表示的点积获得

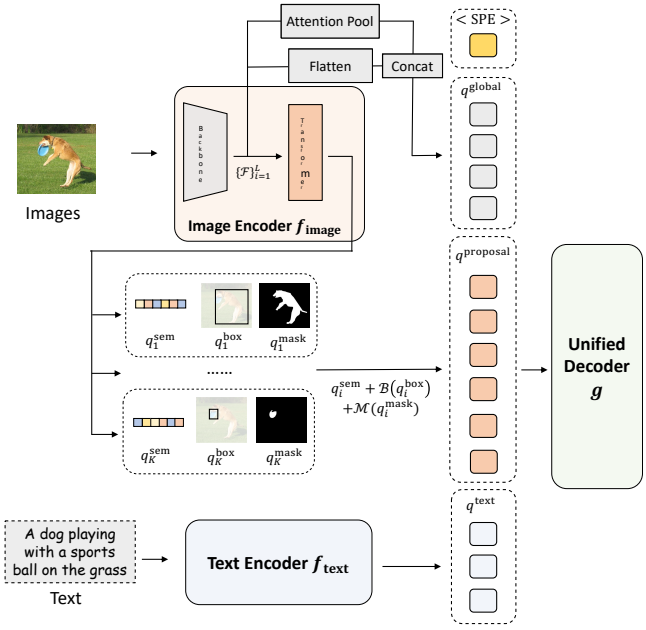


图 3. 我们的 Uni-Perceiver v2 的架构概览。

第 j 个候选方案 q_j 的状态以及逐像素特征图，

$$q_i^{\text{mask}} = \text{Upsample}\Big(\text{MLP}(q_i) \odot \mathcal{R}\big(\mathcal{G}(\mathcal{F}_0) + \mathcal{H}(\mathcal{F}_1^{\text{enc}})\big)\Big), \tag{8}$$

其中，$\mathcal{G}$ 是一个 1×1 卷积层，后接组归一化（GN）[58]；$\mathcal{H}$ 是一个 1×1 卷积，后接 GN 和双线性上采样；$\mathcal{R}$ 是一个 3×3 卷积，后接 GN、ReLU 和一个 1×1 卷积。

区域表示通过融合语义、边界框和分割掩码表示获得。

$$q_j^{\text{proposal}} = q_j^{\text{sem}} + \mathcal{B}(q_j^{\text{box}}) + \mathcal{M}(q_j^{\text{mask}}), \tag{9}$$

其中 $\mathcal{B}$ 表示框坐标的位置编码。$\mathcal{M}$ 使用自适应平均池化将掩码预测缩放到 28×28 的大小。$\mathcal{B}$ 和 $\mathcal{M}$ 都随后经过线性投影以匹配特征维度。注意，在融合之前，边界框和分割掩码表示会被分离。

为了降低计算成本，我们为每个候选区域 q_j^{proposal} 预测其物体性，并选择排名前 O 的候选区域作为最终的区域表示。在 Uni-Perceiver v2 中，O 默认设置为 200。

损失函数。在图像分类等非定位任务中，监督仅施加于统一解码器的最终预测上，如式 7 所示，而对于图像编码器的候选区域生成没有特殊监督。在目标检测等定位任务中，会为区域候选网络的训练施加额外监督。具体而言，我们采用了 con-

trastive query denoising in MaskDINO [17] for the training of the Transformer decoder. For better convergence of the region proposal network, we predict objectness, bounding box, and segmentation mask for each proposal at the outputs of Transformer encoder and each Transformer decoder layer, and detection losses with binary classification (*i.e.,* predicting the objectness instead of classes) are applied to each output as an intermediate supervision.

B. Implementation Details

Region Proposal Network. The hyper-parameters used in our region proposal network are listed in Tab. 7. These values mainly follow Mask DINO [17], but with small modifications. The number of candidate object proposals ('num_queries' in Tab. 7) used to generate regional representations is 300 and 900 for the ResNet-50 backbone and Swin backbones, respectively. To reduce the computation cost of the unified decoder, the region proposals are filtered depending on their objectness scores and only $O = 200$ region representations are selected as the input for the unified decoder ('topk_queries' in Tab. 7). Moreover, to save computation cost, the point loss used in Mask2former [51] is adopted to calculate mask loss, where the number of sampled points is 112×112.

Unified Decoder. As for the Transformer-based unified decoder, a uniform drop rate for stochastic depth is used across all layers and the value is set to 0.1. Unlike Uni-Perceiver series [1, 48], the layer-scale technique [56] is not enabled since the instability phenomenon is not observed when the training of the 6-layers unified decoder. In addition, when Conditional MoE is employed in the unified decoder, the number of experts in each layer is set to 8.

Data augmentation. For all tasks except image detection and segmentation, we apply the data augmentation techniques that are similar to Uni-Perceiver [1]. However, image resolution is set to 384×384 and 224×224 for Swin backbones and for ResNet-50 backbone, respectively. And for object detection and instance segmentation tasks, we first randomly resize the input image with its shorter side between 200 and 1800 pixels and its longer side at most 2400. Then we crop the image to a fixed size of 1600×1600 during training. For evaluation, the shorter side is set to 1400, and the maximum longer side is set to 1600.

Others. Tab. 8 lists the batch size, sampling weight s_k, and scaling factor ω_k for each task and dataset in the joint training.

C. Detection on Novel Categories

Thanks to the general task modeling of Uni-Perceiver v2, different tasks can borrow knowledge from each other. For example, object detection task can generalize to novel categories in image classification dataset. Fig. 4 shows the detection result of Uni-Perceiver v2 on images in ImageNet-1k validation set whose categories do not exist in COCO dataset. This demonstrates the generalization ability of Uni-Perceiver v2, indicating the benefit of general task modeling.

Item	Value
enc_layers	6
dec_layers	6
dim_feedforward	2048
hidden_dim	256
dropout	0.0
nheads	8
num_queries	300/900
topk_queries	200
enc_n_points	4
dec_n_points	4
cls_cost_coef	2.0
bbox_cost_coef	5.0
giou_cost_coef	2.0
mask_cost_coef	5.0
dice_cost_coef	5.0
cls_loss_coef	2.0
bbox_loss_coef	5.0
giou_loss_coef	2.0
mask_loss_coef	5.0
dice_loss_coef	5.0
dn_box_noise_scale	1.0
dn_label_noise_ratio	0.5

Table 7. Hyper-parameters used in our region proposal network.

D. Licenses of Datasets

ImageNet-1k [9] is subject to the ImageNet terms of use [57].
COCO [19] The images are subject to the Flickr terms of use [52].
BookCorpus [49] Replicate Toronto BookCorpus is open-source and licensed under GNU GPL, Version 3.
Wikipedia Most of Wikipedia's text is co-licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported License (CC BY-SA) and the GNU Free Documentation License (GFDL) (unversioned, with no invariant sections, front-cover texts, or back-cover texts). Some text has been imported only under CC BY-SA and CC BY-SA-compatible license and cannot be reused under GFDL.
YFCC [14] All the photos and videos provided in YFCC dataset are licensed under one of the Creative Commons copyright licenses.

MaskDINO [17] 中的对比查询去噪用于训练 Transforme r 解码器。为了使区域提议网络更好地收敛，我们在 Tr ansformer 编码器的输出以及每个 Transformer 解码器层的输出处，为每个提议预测目标性、边界框和分割掩码，并将带有二分类（即预测目标性而非类别）的检测损失应用于每个输出，作为中间监督。

B. 实现细节

区域提议网络。我们的区域提议网络中使用的超参数列于表 7。这些数值主要遵循 Mask DINO [17]，但做了少量修改。用于生成区域表示的候选目标提议数量（表 7 中的“num queries”）对于 ResNet-50 骨干网络和 Swin 骨干网络分别为 300 和 900。为降低统一解码器的计算成本，区域提议会根据其目标性分数进行筛选，并且仅选择 $O = 200$ 个区域表示作为统一解码器的输入（表 7 中的“topk queries”）。此外，为节省计算成本，采用 Mask2former [51] 中使用的点损失来计算掩码损失，其中采样点数量为 112×112。

Item	Value
enc_layers	6
dec_layers	6
dim_feedforward	2048
hidden_dim	256
dropout	0.0
nheads	8
num_queries	300/900
topk_queries	200
enc_n_points	4
dec_n_points	4
cls_cost_coef	2.0
bbox_cost_coef	5.0
giou_cost_coef	2.0
mask_cost_coef	5.0
dice_cost_coef	5.0
cls_loss_coef	2.0
bbox_loss_coef	5.0
giou_loss_coef	2.0
mask_loss_coef	5.0
dice_loss_coef	5.0
dn_box_noise_scale	1.0
dn_label_noise_ratio	0.5

表 7. 我们的区域提议网络中使用的超参数。

统一解码器。对于基于 Transformer 的统一解码器，所有层均采用统一的随机深度丢弃率，其值设为 0.1。与 Uni-Perceiver 系列 [1, 48] 不同，由于在训练 6 层统一解码器时未观察到不稳定现象，因此未启用 layer-scale 技术 [56]。此外，当在统一解码器中采用 Conditional MoE 时，每层的专家数量设为 8。

数据增强。对于除图像检测和分割之外的所有任务，我们采用与 Uni-Perceiver [1] 类似的数据增强技术。不过，对于 Swin 骨干网络和 ResNet-50 骨干网络，图像分辨率分别设置为 384×384 和 224×224。而对于目标检测和实例分割任务，我们首先随机调整输入图像的大小，使其短边在 200 到 1800 像素之间，长边最多为 2400。然后在训练期间将图像裁剪为固定大小 1600 ×1600。评估时，短边设置为 1400，最大长边设置为 1600。

其他。表8列出了联合训练中每个任务和数据集的批大小、采样权重s_k和缩放因子ω_k。

C. 新类别检测

得益于 Uni-Perceiver v2 的通用任务建模，不同任务可以相互借鉴知识。对于例如，目标检测任务可以泛化到图像分类数据集中的新类别。图 4 展示了 Uni-Perceiver v2 在 ImageNet-1k 验证集图像上的检测结果，这些图像的类别不存在于 COCO 数据集中。这证明了 Uni-Perceiver v2 的泛化能力，表明通用任务建模的优势。

D. 数据集的许可证

ImageNet-1k [9] 受 ImageNet 使用条款 [57] 约束。

COCO [19] 这些图像受 Flickr 使用条款 [52] 约束。

BookCorpus [49] 复刻版 Toronto BookCorpus 是开源的，并根据 GNU GPL 第 3 版授权。
维基百科的大部分文本均依据知识共享署名-相同方式共享 3.0 未本地化许可协议（CC BY-SA）和 GNU 自由文档许可证（GFDL）（无版本号，且无不变章节、封面文本或封底文本）进行双重许可。部分文本仅依据 CC BY-SA 及与 CC BY-SA 兼容的许可协议导入，不能在 GFDL 下重复使用。

YFCC [14] YFCC 数据集中提供的所有照片和视频均采用知识共享版权许可之一授权。

task	dataset	#data	batch size / GPU	sampling weight s_k	scaling factor ω_k
Image Classification	ImageNet-1k [9]	1.28M	28	0.1	1.0
Object Detection & Instance Segmentation	COCO [19]	118K	1	0.25	1.0
Masked Language Modeling	Books&Wiki [49]	-	256	0.05	0.5
Image Captioning	YFCC [14]	14.8M	24	0.09831	0.16385
	CC12M [5]	11.1M	24	0.08514	0.1419
	CC3M [33]	3M	24	0.04428	0.0738
	Visual Genome [16]	108K	24	0.02973	0.04955
	COCO Caption [8]	113K	24	0.0192	0.032
	SBU [25]	830K	24	0.02328	0.0388
	sum	29.9M	-	0.3	0.5
Image-Text Retrieval	YFCC [14]	14.8M	28	0.09831	0.3277
	CC12M [5]	11.1M	28	0.08514	0.2838
	CC3M [33]	3M	28	0.04428	0.1476
	Visual Genome [16]	108K	28	0.02973	0.0991
	COCO Caption [8]	113K	28	0.0192	0.064
	SBU [25]	830K	28	0.02328	0.0776
	sum	29.9M	-	0.3	1.0

Table 8. Tasks and datasets used for our joint training. "#data" is the amount of visual training samples. For image captioning and image-text retrieval tasks, a combination of image-text-pair datasets is used for training, which has about 29.9M visual samples after filtering the data overlapping with validation sets. To alleviate the data imbalance problem in the combination of image-text-pair datasets during multi-task training, sampling weight s_k for each dataset is set to be proportional to the square root of the dataset size, which has demonstrated to be effective [48].

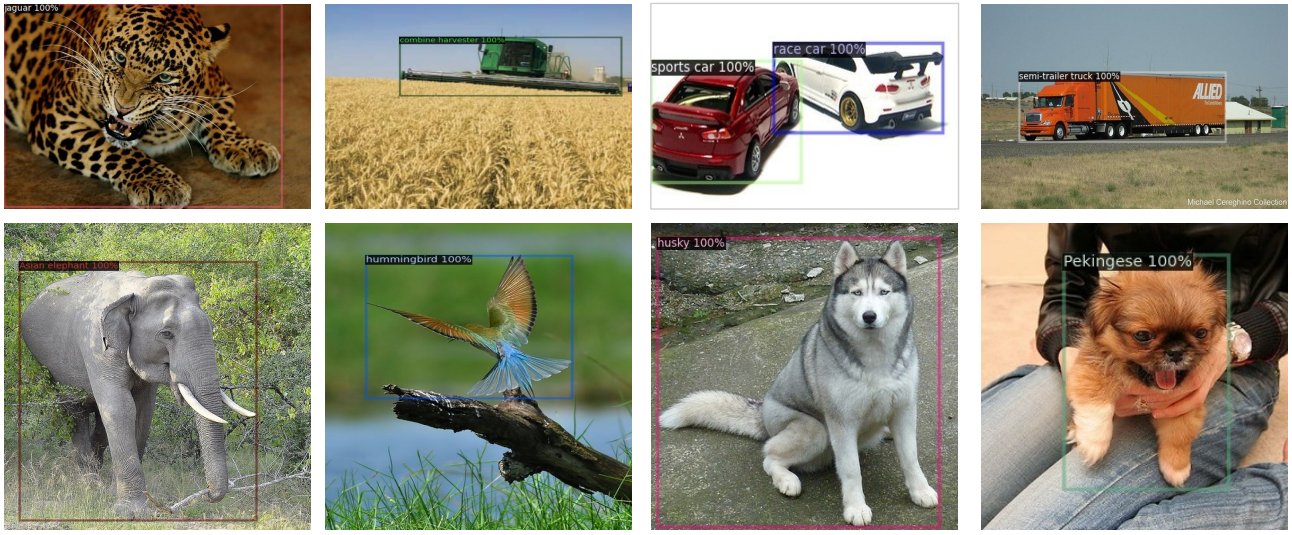


Figure 4. Detection results on novel categories. We show the detection results of images from ImageNet-1k validation set. Note that Uni-Perceiver v2 only uses COCO dataset for the training of image detection task, and most classes in ImageNet-1k are not seen in training.

CC12M [5] is licensed under the Terms of Use of Conceptual 12M [54].
CC3M [33] is licensed under the Conceptual Captions Terms of Use [55].
Visual Genome [16] is licensed under a Creative Commons Attribution 4.0 International License [53].
COCO Captions [8] The images are subject to the Flickr terms of use [52].
SBU Caption [25] The images are subject to the Flickr terms of use [52]

Appendix References

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task	dataset	#data	batch size / GPU	sampling weight s_k	scaling factor ω_k
Image Classification	ImageNet-1k [9]	1.28M	28	0.1	1.0
Object Detection & Instance Segmentation	COCO [19]	118K	1	0.25	1.0
Masked Language Modeling	Books&Wiki [49]	-	256	0.05	0.5
Image Captioning	YFCC [14]	14.8M	24	0.09831	0.16385
	CC12M [5]	11.1M	24	0.08514	0.1419
	CC3M [33]	3M	24	0.04428	0.0738
	Visual Genome [16]	108K	24	0.02973	0.04955
	COCO Caption [8]	113K	24	0.0192	0.032
	SBU [25]	830K	24	0.02328	0.0388
	sum	29.9M	-	0.3	0.5
Image-Text Retrieval	YFCC [14]	14.8M	28	0.09831	0.3277
	CC12M [5]	11.1M	28	0.08514	0.2838
	CC3M [33]	3M	28	0.04428	0.1476
	Visual Genome [16]	108K	28	0.02973	0.0991
	COCO Caption [8]	113K	28	0.0192	0.064
	SBU [25]	830K	28	0.02328	0.0776
	sum	29.9M	-	0.3	1.0

表8. 用于联合训练的任务和数据集。“#data”表示视觉训练样本的数量。对于图像描述生成和图像-文本检索任务，训练时使用了图像-文本对数据集的组合；在过滤掉与验证集重叠的数据后，该组合约包含29.9M个视觉样本。为缓解多任务训练中图像-文本对数据集组合的数据不平衡问题，每个数据集的采样权重s_k被设置为与数据集规模的平方根成正比，这已被证明是有效的[48]。

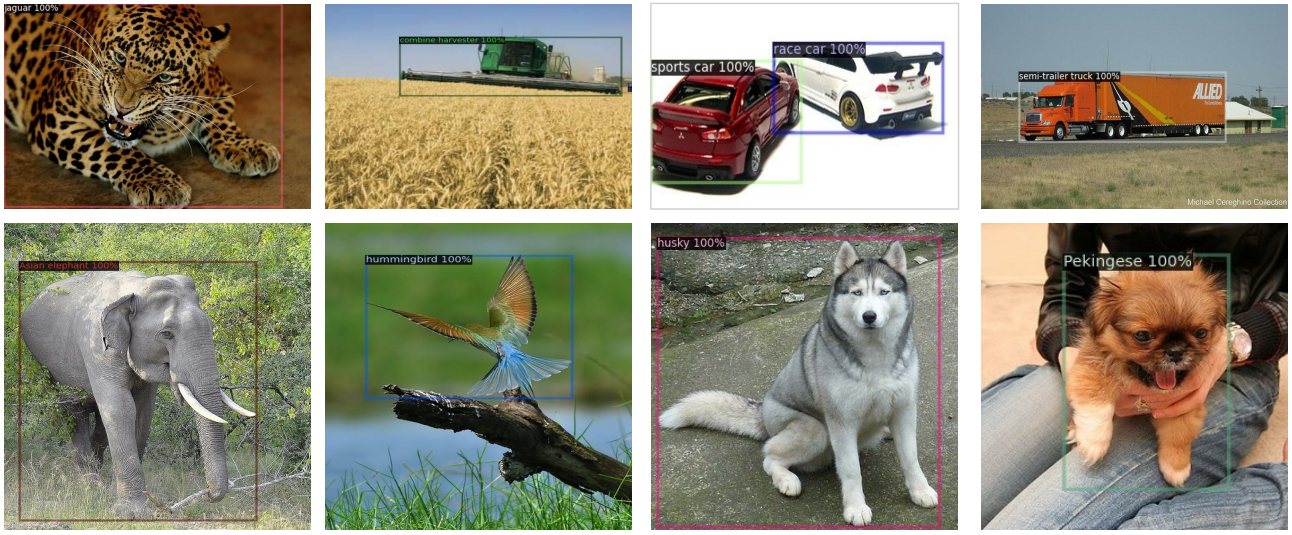


图4. 新类别上的检测结果。我们展示了 ImageNet-1k 验证集图像的检测结果。请注意，Uni-Perceiver v2 仅使用 COCO 数据集进行图像检测任务的训练，而 ImageNet-1k 中的大多数类别在训练中并未出现。

CC12M [5] 根据 Conceptual 12M [54] 的使用条款授权。

CC3M [33] 根据 Conceptual Captions 使用条款 [55] 获得许可。

Visual Genome [16] 采用知识共享署名 4.0 国际许可协议 [53] 授权。

COCO Captions [8] 图像受 Flickr 使用条款 [52] 约束。

SBU Caption [25] 这些图像受 Flickr 约束使用条款 [52]

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